A BIOINSPIRED ECO-FRIENDLY PLANT PROTEIN-BASED POROUS NANOFIBROUS AEROGELS FOR MULTIFUNCTIONAL APPLICATIONS IN WATER PURIFICATION AND OIL/WATER SEPARATION

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Certificate of originality

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Abstract

Oil-absorption plays an important role in many fields such as oil-spill cleanup, oil/water separation, and environmental remediation. Physical absorption using absorbent materials is generally considered to be one of the most effective countermeasures because it is simple, efficient, and fast. Porous materials with hydrophobicity are good candidates for oil absorption because the porous structure helps them to absorb oil in the presence of water. While hydrophobicity and oleophilicity are primary determinants of successful sorbents, biodegradability is also an important factor. Natural materials are biodegradable and environment friendly, avoiding secondary pollution. Corn zein is an abundant and reproducible biopolymer with excellent biodegradability and hydrophobicity in specific conditions. In this study, the bio-inspired zein scaffolds and gelatin/zein fibers were fabricated to obtain enhanced properties for applications in water purification and oil-water separation.

In Chapter 3, zein nanofibers were fabricated using a facile and cost-effective method. Bio-inspired by the natural spider silk fibers, the scaffolds were formed from the dispersion of zein nanofiber in tert-butanol and their oil absorption capacities were investigated. Various fiber to solvent ratios, 1:0, 1:150, 1:125, 1:100, 1:75, were selected and the scaffolds, E0, E150, E125, E100, and E75, respectively, were obtained after freeze-drying. The zein scaffolds as well as the spider silk fiber scaffold (SSFS) were characterized using attenuated total reflection-fourier transform infrared spectroscopy (ATR-FTIR), water contact angle (WCA) goniometer, and scanning electron microscopy (SEM). The WCAs of the scaffolds E75, E100, E125, E150, E0, and SSFS were 106.120°, 100.233°, 103.125°, 98.265°, 110.743°, and 106.674°, respectively, which indicated that they were hydrophobic. The scaffolds exhibited enormously high absorbing capacities for organic liquids. The absorption

capacity of the E0 in motor oil, diesel oil, pump oil, and olive oil was 91 g/g, 152 g/g, 77 g/g and 147 g/g respectively. The oil absorption capacity of the SSFS in motor oil, diesel oil, pump oil, and olive oil was 53 g/g, 29 g/g, 52 g/g and 31 g/g respectively. It was lower than that of the E0, but higher than that of most oil absorbents. The porosities of E150, E125, E100, E75 were much higher than that of E0 but the oil absorption capacities of these scaffolds were lower than that of E0. Hence, porosity has limited influence on oil absorption capacities and the organic solvent absorption capacity is related to the density of solvent. E0 and SSFS had good reusability for 30 cycles while other scaffolds were broken in absorption–squeezing cycles. It is thought that the formation of hydrophobic surface was mainly contributed by the enriched β-sheet content. E0 has similar molecular structure, hydrophobicity and oil absorption capacity to SSFS.

In Chapter 4, gelatin/zein nanofibers at various ratios in two different solvents 80% acetic acid and solution mixed by water/ethanol/acetic acid in the ratio of 6/2/2 were fabricated using electrospinning. The nanofibers exhibited excellent organic liquids absorption capacities with the highest oil absorption 109 g/g for olive oil, 48 g/g for diesel oil and 166 g/g for motor oil, approximately 4-6 times that of natural sorbents and nonwoven polypropylene fibrous mats. The morphology, diameter distribution, and porosity of the nanofibers were investigated. Their chemical structure, mechanical strength, hydrophobicity and thermal property were characterized using ATR-FTIR, mechanical test, water contact angle measurements and thermogravimetric analysis. The hydrophobicity of the fibers was increased by the increase of gelatin/zein and obtained the highest WCA (114.3°) with GZ ratio equals to 1:1. With a further increasing of zein or gelatin, WCAs were decreased. The gelatin/zein nanofibers in 80% acetic acid with weight ratios of 1:3 and 1:2 showed much higher elongation at break of 16.4% and 11.0%, according to mechanical tests. It indicated these nanofibers had good deformability and flexibility. The gelatin/zein composite nanofibers were also in high potential for emulsion separation and heavy metal ion absorption. This work suggested that the zein based scaffolds and composite gelatin/zein nanofibers have potentials as materials for oil absorption and filtration applications.

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Abbreviations

Carbon nanotube: CNT

Circular dichroism: CD

Melamine sponge: MS

Polyurethane: PU

Scanning electron microscopy: SEM

Spider silk fiber scaffold: SSFS

Tert-butanol: TBA

Thermogravimetric analysis: TGA

Three-dimensional: 3D

Total reflection-fourier transform infrared spectroscopy: ATR-FTIR

Two-dimensional: 2D

Water contact angle: WCA

Water sliding angle: WSA

Chapter 1: Literature Review

1.1 Oily wastewater

The origins of oily wastewater generally result from petrochemical, heavy metal food processing, metal cuttings, paint, edible oil producing industries and automobile industrial premises (Wuana & Okieimen, 2011). Kitchen waste, oil spills of mechanical devices and human activities are also potent resources for oily wastewater generation (Jamaly, Giwa, & Hasan, 2015). The incidence of oil-spill is also the main source of oily wastewater and causes serious environment and economy issues (Wahi, Abdullah, Shean, Choong, & Ngaini, 2013). The important major oil spills are summarized in Table 1. This oil-polluted wastewater is steadily increasing seriousness of the environment, mutagenic and hazardous to human health and inhibitory to plant growth. The disposal of oily water stream into water bodies may also impart a sunlight impervious layer on the surface of the stream (Putatunda, Bhattacharya, Sen, & Bhattacharjee, 2019). It causes the enhancement of sunlight followed by disruption of an aquatic ecosystem. Therefore, it's one of the primary concerns to apply an efficiency and proper technology of oily wastewater before its expulsion to the environment.

Researchers have used several technologies such as, in situ burning (Mullin & Champ, 2003), dispersant treatment (Lessard & Demarco, 2000), physical adsorption (Carmody, Frost, Xi, & Kokot, 2008), chemical coagulation (Putatunda et al., 2019), and bioremediation to purify this oil-contaminated wastewater to the accepted level (José et al., 2019). However, these techniques have certain toxicity, accompanying expenses, and programs requiring tedious and time-consuming procedures. These drawbacks limit their practical application. Especially, the design of oil-containing wastewater emulsion technology is complex. Oil and water become completely miscible to form an emulsion in the presence of surfactants and ions (Bhattacharya & Bhattacharjee, 2018). Therefore, exploring treatments that can solve the water purification and oil-water separation problems becomes a major task to achieve.

Table 1: List of import major oil spills of the world (Of, 2018).

No	Oil spill	Year	Location	Quantity (tones)
1	Torrey Canyon	1976	Scilly Isles, UK	119000
2	Sea Star	1972	Gulf of Osman	115000
3	Jakob Maersk	1975	Oporto, Portgal	88000
4	Urquiola	1976	La Coruna, Spain	100000
5	Hawaiian Patriot	1977	300 nautical miles off Honolulu	95000
6	Amoco Cadiz	1978	Off Brittany, France	223000
7	Atlantic Empress	1979	Off Tobago, West Indies	287000
8	Independenta	1979	Bosphorus, Turkey	94000
9	Ixtox I oil wel	1979	Gulf of Mexico	454000
10	Irenes Serenade	1980	Navarino Bay, Greece	100000

11	Abt Summer	1991	700 nautical miles off Angola	260000
12	Castillo De Bellver	1983	Off Saldanha Bay, South Africa	252000
13	Nova	1985	Off Kharg Island, Gulf of Iran	70000
14	Odyssey	1988	700 nautical miles off Nova Scotia, Canada	132000
15	Khark 5	1989	120 nautical miles off Atlantic coast of Morocco	70000
16	Exxon Valdez	1989	Prince William Sound, Alaska, USA	37000
17	Haven	1991	Genoa, Italy	144000
18	ABT Summer	1991	700 miles of the Angolan coast	260000
19	Aegean Sea	1992	La Coruna, Spain	74000
20	Katina P	1992	Off Maputo, Mozambique	67000
21	Fergana valley	1992	Fergana valley of Central Asia	285000
22	Braer	1993	Shetland Island, UK	85000
23	Sea Empress	1996	Milford Haven, UK	72000
24	Prestige	2002	Off Galicia, Spain	63000
25	Hebei Spirit	2007	Taean, Republic of Korea	11000
26	2014 Israeli Oil spill	2014	Trans-Israel Pipeline, Israel	1948-4300
27	Refugio oil spill	2015	California Coast, USA	330

1.2 Organic compounds in oily wastewater

The organic compounds in oily-wastewater are categorized into four groups based on their chemical structures: aliphatic, aromatics, nitrogen sulfur oxygen (NSO) containing compounds and asphaltenes (Mrayyan & Battikhi, 2005; Reddy, Devi, Chandrasekhar, Goud, & Mohan, 2011). Generally, organic compounds in oily-wastewater include alkanes, cycloalkanes, benzene, toluene, xylenes, naphthalene, phenols, and various polycyclic aromatic hydrocarbons (PAHs) (e.g., methylated derivatives of fluorine, phenanthrene, anthracene, chrysene, benzofluorene, and pyrene) (Padaki et al., 2015).

1.3 Harmful effects of oily wastewater in environment and human beings

First, oily wastewater pollution affects the resources of drinking-water, groundwater, and endangering aquatic resources. It may also cause atmospheric pollution and destructing the natural landscape. Toxic compounds such as phenols, petroleum hydrocarbons, and polyaromatic hydrocarbons in oily wastewater are thought to inhibit the growth of plant and animal (Parus et al., 2017). Some even believe that oil may be toxic to certain microorganisms, which in turn are responsible for bioremediation of wastewater (Wahi et al., 2013).

Moreover, they are harmful to human beings. The proliferation of the toxic components may cause marine pollution through marine ecology and food chain in the affected marine environment. It was even found that the consumption of oil-contaminated marine food might cause detrimental effects on the health of human beings. Oil compounds are bio-accumulated so it’s necessary to separate oily contaminants from water before their discharge (Aguilera, Méndez, & Laffon, 2010). It was found that oily wastewater could spread its toxicity not only in the way of direct consumption of pollutants, but also by direct contact. Due to the carcinogenic properties of hydrocarbon oil exposure, long-term exposure to hydrocarbon oil wastewater could also increase the risk of developing skin tumors (Wahi et al., 2013).

1.4 Oil spill clean-up technologies

Currently, there are four groups of methods for oil spill clean-up: physical treatments including mechanical recovery and membrane separation (Mullin & Champ, 2003), chemical methods , and bioremediation. Mechanical recovery includes booms, skimmers, oil-waster separators and adsorbents, which are expensive and need specialist personnel and equipment. Chemical methods including dispersion, in-situ burning and solidification, are expensive and unfavorable to the environment (Page, Bonner, Mcdonald, & Autenrieth, 2002). These reagents can be harmful to the environment, which would cause the limitation in their applications. Microbial degradation of hydrocarbons in oil is an effective and time-consuming bioremediation method. Its effectiveness is significantly affected by temperature, oxygen content and organic species (Atlas, 1995; Prince, 1997). Therefore, their efficiency of removing residual oil is quite limited.

1.4.1 Physical treatments

Mechanical recovery

Mechanical recovery is usually based on using barriers like booms, skimmers and adsorbents to control the spread of oil (Mullin & Champ, 2003). Booms are used to enclose the oil to prevent its spreading and remove oil from biologically sensitive areas (Journal, Sciences, & Publications, 2011). Skimmers are mainly categorized into oleophilic, suction and weir skimmers. Skimmers are usually used with booms to remove floating oil from the water surface. Floating is an accelerated gravity separation technology, in which tiny bubbles are injected into the water phase containing oil or oily solid particles. These bubbles attach themselves on the droplets and make the oil lighter. Then the density difference between water and oil agglomerate is increased so the oil droplets and oil-coated solids rise to the surface in the form of foam. In this way, they can be effectively separated from the aqueous phase by the flotation chamber (Moosai & Dawe, 2003). The most commonly used floating includes flotation dissolved air flotation, flotation and jet impeller flotation methods (Yu & Han, 2017). Table 2 depicts the mainly used floating treatment of oily wastewater. Adsorbents including natural organic, natural inorganic and synthetic materials which are used to facilitate the conversion oil to a semi-solid phase after skimming.

Mechanical recovery is the primary response to oil spill cleanup with efficiency up to 90%, but operations for large volumes of leaks are often expensive, complex, and labour-intensive (Yavari & Malakahmad, 2015). Each recovery operation involves two or more large ships, trained rescue workers, hundreds of meters of containment, skimming systems, pumping equipment, temporary storage and disposal systems. It is generally not possible to perform the closing and recycling operations at speeds higher than one or two sections. Besides, mechanical recovery produces a mixture of oil, water and debris that need extra work like recycling, landfill or incineration. The efficiency of mechanical recovery is low that rarely exceeds 20% (Bhattacharya & Bhattacharjee, 2018). Scientists are trying to improve the efficiency by appropriately utilizing the window of opportunity for a particular oil spill combined with its weathering and emulsification rates.

Table 2. Mainly used floating treatment of oily wastewater (Yu & Han, 2017).

Flotation type	Oil treatment efficiency
Flotation	> 90%
Peeling flotation	81.4%
Dissolved air flotation	92.5%
Dissolved air flotation	> 90%

1.4.2 Membrane separation technology

Membrane separation technology is a method to separate two phases based on adsorption, sieving and electrostatic phenomenon (Padaki et al., 2015). The mechanism of adsorption is correlated with the hydrophobic/hydrophilic interaction of solute and membrane. Because of this interaction, the pore size is reduced leading to more rejection. The membrane separation processes can be divided into microfiltration (50-500 nm), ultrafiltration (2-50 nm), nanofiltration (less than 2 nm), and reverse osmosis (0.3-0.6 nm) according to the pore size (Padaki et al., 2015). The membrane separation technology is in low energy requirement, less pollution, easy to handle and high efficiency as compared to conventional techniques. However, the development of the novel and economic performance of film to overcome some shortcomings (like thermal stability, resistant to corrosion, the film is likely to be contaminated and the process having small volume) is in high demand. The oily wastewater treatments by membrane separation technology are presented in Table 3.

Table 3. The oily wastewater treatments by membrane separation technologies (Cui, Zhang, Liu, Liu, & Lun, 2008; Hua et al., 2007; Yang, Ma, & Yang, 2011).

Membrane separation technologies	Removal efficiency
UF	Oil content < 1 mg/L
Microfiltration	97%
Dynamic membrane	99%
Microfiltration	92.4% (TOC removal)
Microfiltration	99%
Nano-porous membrane	76.9% (COD removal)
Nano-porous membrane-powdered activated carbon	71.5% (TOC removal)

1.4.3 Chemical treatments

Dispersion, in-situ burning and solidification are three main chemical ways to counter the effects of oil-spill. Dispersion is the way that uses dispersants to break down the oil into smaller droplets. It makes the oil much easier for oil degradation by natural processes (Lessard & Demarco, 2000). Dispersants are detergent-like products made of surfactants dissolved in one or more solvents. It includes biosurfactants (aescin, lecithin, rhamnolipid, saponin and tannin) and chemical surfactant (sodium dodecyl sulphate, and SDS) (Urum & Pekdemir, 2004). The surfactants could diffuse to the oil/water interface because they are designed to be lipophilic and hydrophilic. When applied to a film of oil, they align themselves to attach to the oil phase by the lipophilic end of the molecule and extend into the water phase by the hydrophilic end. In this way, the interfacial surface tension between water and oil is reduced and oil is dispersed into the water column at very low concentration (Lessard & Demarco, 2000). The chemical dispersion mechanism is shown in Figure 1.

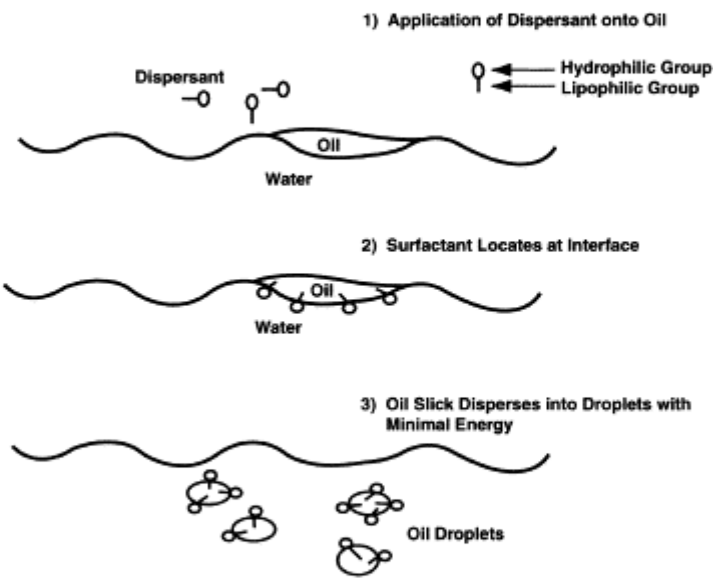


Figure 1. The chemical mechanism of dispersion (Lessard & Demarco, 2000).

Dispersion is increasingly supported because they are low in toxicity, effective in a wider range of conditions such as rough seas, strong winds and ocean currents, which prevent the formation of water-in-oil emulsions and allow for rapid treatment of large areas. Generally, the efficiency of biosurfactants like lecithin aescin, and fatty acyl glutamate is 42%, and chemical surfactant like sodium dodecyl sulfate could get much higher efficiency up to 80%. However, this is time-consuming process because it takes time for the chemical agents to penetrate the oil and reach the interface. Although the toxicity is low, if dispersants exist in the water column at a rate of several hundred parts per million for more than several days, it would also produce lethal or sub-lethal toxicities.

In situ burning is the technique using the controlled ignition and burning of the oil at or near the spill site (Mullin & Champ, 2003). Burning is an effective way to remove oil from the water surface when it is used before the oil weathers and releases its volatile components. The removal efficiencies are easily exceeded by 95% (Mullin & Champ, 2003). In situ burning is believed to be logistically simple, rapid, inexpensive and less complex than conventional operations which involved in mechanical recovery, transfer, storage, treatment and disposal. However, successful burn operations are limited by environment (extremely low temperatures, ice-infested waters), weather and oil properties (flame temperatures, slick thickness, oil type and so on) (Mullin & Champ, 2003). It is hard to apply when oil formed a stable emulsion because it is unignitable when water content exceeds 25%. Safety issues should also be considered.

Solidification is the method that uses solidifier with a strong affinity for oil rather than water to form physical bonding with oil and turn oil into a solid rubber state. Then it can be easily removed by physical methods (Of, 2018). Solidifiers are mainly three types: polymer sorbents, cross-linking agents, and polymers with cross-linking agents (Fingas, 2012). The operations of solidification are simple but the toxicity of both the solidifier and the solidified oil is an important issue.

1.4.4 Biological treatments

Biological treatments are based on using organisms mainly microorganisms and plants for remediation. Sometimes, natural remediators are used to prevent further damages to the environment. The generally used microorganisms (fungi and bacteria) and phytoremediators are listed in Table 4. Bioremediation is environment-friendly and in high efficiency (about 80%) but the limitations of microbial activity in the soil restricted the application (Atlas, 1995). Microorganisms have a high requirement of growth condition and it is also possible to produce toxic metabolites (Nikolopoulou, Pasadakis, & Kalogerakis, 2007).

Table 4: Generally used fungi bacteria, and phytoremediators in bioremediation (Ibrahim, Lateef, Khalifa, & Monem, 2013; Li et al., 2014; Lotfy & Mostafa, 2014; Of, 2018; Thi, Ha, Sakakibara, & Sano, 2011).

No	Microorganisms		Phytoremediators
	Fungi	Bacteria	
1	*Aspergillus sp.*	*Arthobacter sp.*	*Medicago sativa*
2	*Candida sp.*	*Brevibacterium sp.*	*Lolium arundinaceum*
3	*Fusarium sp.*	*Dietzia sp.*	*Lolium multiforum*
4	*Trichoderma sp.*	*Nocardia sp.*	*Impatiens balsamina*
5	*Phanerochaete sp.*	*Pseudomonas sp.*	*Canna indica*
6	*Mortierella sp.*	*Rhodococcus sp.*	*Chromolaena odorata*
7			*Cyperus rotundus*
8			*Biden pilosa*
9			*Iris dichotoma Pall. And Iris lacteal Pall.*

10	*Ludwigia octovalvis*
11	*Eleocharis acicularis*
12	*Callitriche lusitanica*
13	*Fontinalis antipyretica*
14	*Callitriche stagnalis*
15	*Helianthus annuus*
16	*Zea mays*
17	*Iris pseudacorus*
18	*Lemna minor & Spirodela polyrhiza*
19	*Medicago sativa*
20	*Tagetes patula*
21	*Populus deltoids* × *Populus nigra*

1.5 Oil absorbents

It is one of the most economical and efficient approaches for oil-water separation using mechanical extraction by absorbents. The absorbents are used to concentrate and transform oil from the spilt area to the semisolid or solid-phase (Wang, Yu, Sun, & Ding, 2016). It has drawn significant attention that porous materials are used as oil absorbents because they can separate oil from water in a simple, fast, and strong absorption process. Generally, an ideal absorbent should have a strong oil absorption capacity, high oil-water selectivity, low density (high floatability), excellent recyclability and no-toxicity. There are three kinds of porous absorbents in the process of oil remediation, namely natural products, synthetic organic products, and inorganic mineral products (Adebajo, Frost, Kloprogge, Carmody, & Kokot, 2003). Table 5 summarized the oil absorption performance of various porous oil absorbents. The main limitations are their poor removal ability, selectivity, and slow kinetics. Hence, it is necessary to find the better alternative absorption materials with high oil absorption capacity and high uptake rate.

Various reusable oil absorbents have been fabricated on the basis of natural materials as well as synthetic origins (Moosai & Dawe, 2003; Yu & Han, 2017). Natural absorbents include fibrous absorbents such as wool, cellulose, cotton, sisal, hay, feather, and straw and mineral absorbents such as perlite, clay, peat moss, and vermiculite (Reddy et al., 2011; Yavari & Malakahmad, 2015). Synthetic absorbents include polyurethane (PU), rubber sponges, nonwoven polypropylene mats, silicon aerogels, metal-organic framework mesh, and modified activated carbons (Ibrahim et al., 2013). The natural absorbents in actual applications have some limitations like less efficiency, easy exhaustion, and poor recycling performance. Different techniques have been used to prepare artificial absorbents. However, the application of these approaches are limited by expense and toxicity and the possibility of serious risks to human health and the environment.

1.5.1 Carbon-based aerogels

Carbon-based aerogels could be divided into carbon nanotube (CNT), carbon fiber, carbon micro-belt, and graphene aerogel. The main building blocks of carbon-based aerogels include biochar (Lee et al., 2018), carbon fibers (Hua et al., 2012), carbon nanotubes (CNTs) (Gui et al., 2010) and graphene. These 3D structures materials have excellent properties like ultra-low density, high porosity, large specific surface area, and super-hydrophobicity (Gui et al., 2013; Lei et al., 2013; Zhang et al., 2013). For example, aerogels based on twisted carbon fibers have the ability to absorb a large number of organic compounds and oils with a maximum absorption capacity of 192 g/g (Bi et al., 2013). The porosity structure of CNT aerogels has a positive influence on oil/organic absorption capacity. It was also found that the CNT aerogels with a porosity higher than 99% could absorb numerous organics and oils with remarkable selectivity and absorption capacities up to180 g/g (Chem et al., 2011). However, the precursors of carbon aerogels are expensive and environmentally risky, the implementation set up is complex, and the mass productions are less feasibility. These drawbacks are less attractive for their applications (Bi et al., 2013).

Based on this consideration, an eco-friendly, cost-effective and biosustainable carbon aerogel has been produced from a crude biomass sources derived from vegetables (Wei et al., 2013; Y. Li et al., 2014). The winter melon (wax gourd) is considered to be a fast-growing long-season vegetable and it holds 90% water and about 10% polysaccharides. In the process of robust hydrothermal carbonization under mild heating conditions, it can be easily converted into the porous carbon materials (Y. Li et al., 2014). Watermelon carbon aerogels with low density and hydrophobicity were able to selectively absorb a broad range of organic solvents with a maximum oil absorption capacity 16–50 g/g and over 5 absorption–harvesting cycles (Y. Li et al., 2014). A lightweight, hydrophobic and porous carbon aerogels are developed by the waste newspaper, a novel raw carbon source. It is made through freeze-drying and post-pyrolysis processes. The carbon aerogels developed in this way showed an excellent hydrophobic property with an uptake capacity of 29–51 g/g towards a great number of organics and oils (Han et al., 2016).

1.5.2 Cellulose aerogel

Cellulose aerogel (Nguyen et al., 2013 is the aerogel composed of eco-friendly cellulose fibers. The porosity and surface area of these aerogel are quite high (Feng et al., 2015). There are abundant cellulose fiber sources from woods, plants, algae, bacteria (Amor et al., 1995; Chen et al., 2011; Sai, Fu and Li, 2015) and the price of these raw materials is low. Therefore, the cellulose aerogels have become a very promising candidate to apply in oil absorption. This biodegradable aerogel is considered as a novel type of synthetic organic absorbent with an absorption capacity around 20 g /g (H. Liu et al., 2017). Recently, the recycled paper wastes were fabricated into a lightweight and highly porous cellulose aerogels using an alkaline/urea processing method (Nguyen et al., 2013). The waste newspaper encompassing 7% public solid wastes is considered to be a normal waste biomass and a cheap cellulosic material. The recycled cellulose aerogel is hydrophilic but after coated by a hydrophobic surface it has the ability of exhibiting higher absorption capacities of 24.4 g/g for crude oils. To enhance the absorption capacity, a fast synthesis procedure for the development of cellulose aerogel from the waste papers was developed. A Kymene cross-linker was used to promote the gelation process replacing the role of an alkaline/urea reagent (Feng et al., 2015). In the process of gelation, Kymene cross-linker formed a hydrogen bonding between the cellulose molecules and performed a cross-linking reaction by itself. Therefore, the recycled cellulose aerogels obtained robust and flexible features. After coated by methyltrimethoxysilane, the hybrid recycled cellulose aerogel become superhydrophobic and superflexible. It showed a high degree of porosity and possessed a mesoporous characteristic (size of pores <50 nm). The recycled cellulose aerogels showed a maximum absorption capacity of 95 g/g in motor oil with the 0.25 wt.% cellulose aerogel. It was mainly due to its lowest density (0.007 g/cm^3) and highest porosity (99.4%). Additionally, the viscosities of the oil used, the porosities of these absorbents and the environmental pH could influence their absorption capacity.

1.5.3 Composite aerogel

To overcome the drawbacks of normally aerogels and get a better application in oil-spill cleanup, composite aerogels were fabricated (Sai et al., 2014). Composite aerogel was also used for removing these toxic organic compounds and oil spills. It has better mechanical compressibility and hydrophobicity the synthesis of its components. In this consideration, a composite aerogel made by the interpenetrated network of silica aerogel and bacterial cellulose were fabricated. The surface area was as high as 734.1 m^2 /g, high contact angle was up to 133° and the improved modulus of elasticity approached to 4.15 MPa (Sai et al., 2014). The prepared composite aerogels indicated an outstanding oil uptake capacity with absorbing 400 ml of vegetable oil by 60 mg aerogel composites (Sai et al., 2014).

1.5.4 Sponges

Commercial oil absorption sponges including polyurethane (PU) sponge and melamine sponge (MS), have widely developed in the field of oil-water separation. They are generally considered to be low cost, high porosity, flexible design and 3D skeleton structures. However, the amphipathicity of both PU sponge and MS restricts their application in oil-water separation. Therefore, recent studies have focused on the surface modification of sponges to obtain a hydrophobic surface. Materials used to decorate the sponges include silanes (Pham and Dickerson, 2014), hydrophobic polymers (Online et al., 2013; Zhou et al., 2013), hydrophobic nanoparticles (Feng et al., 2015; Lei et al., 2017), carbon materials (He et al., 2018) and so on. The coating material usually uses organosilicons because of its hydrophobic nature. Organosilanes are commonly selected to decrease the surface energy of the sponge surface. They include vinyltrimethoxysilane (VTMS) (Gao et al., 2018), octadecyltrichlorosilane (Ke et al., 2014; Pham and Dickerson, 2014), methyltrichlorosilane (Tahara et al., 2013; Peng et al., 2019), heptadecafluoro-1,1,2,2-tetrahydrodecyltrimethoxysilane (FAS-17) (Liu et al., 2015), hexadecyltrimethoxysilane (Yeom and Kim, 2016), fluoroalkylsilane Zhang et al., 2013), and dodecafluoro heptylpropyl-trimethoxylsilane (Actyflon-G502) (L. Liu et al., 2017). Methyltrichlorosilane salinized sponge with a WCA of 143° exhibited an absorption capability of 65 g/g in diesel (Peng *et al.*, 2019). Hydrophobic polymer is widely used to enhance the hydrophobicity of sponge. PU sponge grafted by a layer of polymer molecular brush exhibited higher hydrophobicity with WCA of 152° and oil absorption capacity up to 40.91 g/g. The sponge could obtain rough and hydrophobic surface by attaching hydrophobic particle materials on its smooth surface. Particles which are widely used include TiO_2 (Lei *et al.*, 2017), SiO_2 (Yeom and Kim, 2016), Ag (B. Li *et al.*, 2014), ZnO (Tran and Lee, 2017), Fe_3O_4 (Li *et al.*, 2018), attapulgite particles (Manuscript, 2016), kaolinite (Peng *et al.*, 2018), and so on. TiO_2 nanoparticles modified sponge exhibited superhydrophobic property with WCA of 161° and absorption capacity up to 88.1 g/g (Lei *et al.*, 2017). Fe_3O_4/PU sponge (Li *et al.*, 2018) exhibited good water repellency with WCA of 154.7° and absorption capacity up to 43 g/g. Carbon materials are widely used in the field of oil spill treatment owing to their intrinsic hydrophobicity. Nano-diamond is one of the most promising new carbon-based materials and has a wide range of applications. Nano-diamond modified PU sponge shows the superhydrophobic property (WCA =160°), so it has good selectivity and adsorption capacity (15-60 g/g) (Cao *et al.*, 2016).

1.6 Hydrophobicity and oil absorption

Hydrophobicity is one of the most important factors influencing the oil absorption capacity. Generally, highly hydrophobic absorbents show more efficient oil absorption capacity than that of low-hydrophobic absorbents. Absorbents with highly hydrophobicity and porous structures could be a very efficient tool of oil-spill removal. For this purpose, scientific research and industrial developments are trying to get a novel eco-friendly and superhydrophobic oil-water separation materials recently (Xue *et al.*, 2011). From the environmental and economic perspectives, it is a smart approach for oil-spill remediation and selectively oil separation using well-designed water-repellant superhydrophobic materials. The operation is considered to be simple, and the materials could be reused for multiple times.

1.6.1 Theoretical basis on surface wettability

When the surface of an absorbent material simultaneously possesses microscale and nanoscale roughness, a water droplet can effectively be repelled by air pockets that exist between the water droplet and the material surface. Surface wettability can be evaluated by testing the contact angle (CA) and sliding angle (SA) of a small liquid droplet on the target substrate. The CA is generally used to describe the static aspect of surface wettability while the SA can reflect the dynamic behavior of a droplet on the substrate (Chen and Yang, 2017; Liu, Wang and Jiang, 2017; Manuscript, 2017). For a droplet on a material surface, CA refers to the angle between the tangent to the air/liquid curved surface and the liquid/solid interface at the three-phase contact line. A hydrophilic substrate is defined as water droplet with a water CA (WCA) smaller than 90° while a hydrophobic substrate is defined as a WCA larger than 90°. A substrate with WCA less than 10° or WCA larger than 150° are defined as superhydrophilicity and super-hydrophobicity, respectively. Such materials are also potential candidates as effective oil absorbents (Choi *et al.*, 2011; Song *et al.*, 2014). The combination of tunable surface wettability and low surface energy is a promising research direction for engineering porous materials for effective oil-water separation (Halake *et al.*, 2018). In most cases, the substrate surface usually has a different degree of roughness rather than ideal smoothness. The rough surface microstructures also have a great influence on the wettability of the materials besides their chemical composition (Wang and Jiang, 2007). The wettability of a liquid droplet on a rough substrate can be described by three wetting models: Wenzel state, transition state, and Cassie state (Wang and Jiang, 2007; Wang *et al.*, 2015; Chen and Yang, 2017; Manuscript, 2017).

1.6.2 Oil/water separation by the superwetting porous materials

It is necessary to achieve effective and repeatable oil-water separation by extreme wettability and porous microstructure. Superhydrophobic porous membranes, underwater super-oleophobic porous membranes, and superhydrophobic 3D porous oil absorbent are three most commonly used superwetting materials in highly efficient oil-water separation (Article and Key, 2014; Manuscript, 2016; Gupta *et al.*, 2017). The oil-water separation process by superhydrophobic porous membrane (such as metal meshes, fabrics, fibers, polymer membranes and so on), underwater super-oleophobic porous membranes, and superhydrophobic 3D porous oil absorbent is displayed in Fig. 2. Driven by super-oleophilicity and gravity force, the oil phase in the mixture can wet and penetrate through the membrane. Because of the super-hydrophobicity of the membrane surface, the water phase is prevented from passing through this membrane (Yu *et al.*, 2014). As a result, oil is removed from the mixture and separated from the water. According to the formation principle of underwater superoleophobicity, the surfaces of underwater super-oleophobic porous membranes usually show superhydrophilicity (or strong hydrophilicity) in air while the superoleophobicity in water. Once the oil-water mixture is poured onto such membrane which is previously wetted by water, the water in the mixture to permeate through the membrane leading by the superhydrophilicity. On the contrary, oil always stays on the membrane. Because of underwater superoleophobicity, the oil phase is repelled by the pre-wetted membrane (Yong *et al.*, 2016). Therefore, water in the mixture is removed, and only oil left. The superhydrophobic 3D porous oil absorbent (such as sponge, foam, aerogel and so on) can directly absorb and remove oils from water in an absorption way, and it is displayed in Figure 2c. In this process, oil phase of the mixture could be absorbed and enter into the inner space of these 3D porous materials due to super-oleophilicity and capillary action. On the contrary, the water phase is repelled because of the superhydrophobic surfaces of these materials. In this way, oil phase is selectively absorbed by the superhydrophobic 3D porous materials (Song *et al.*, 2014). The oil part could be removed from the mixture by taking the materials out of the mixture liquid, and the absorbed oil can be easily released and collected by squeezing. Particularly, oils both on water surface and underwater could be removed by the superhydrophobic 3D materials. However, the oil-water mixture is often varied and the reported superwetting materials and the separating ways used in oil-water separation are not limited to the above-mentioned three types (Article and Key, 2014; Manuscript, 2016; Gupta *et al.*, 2017). Hence, the separating materials should show their respective characteristics in structure and wettability, according to the different types and properties of the oil-water mixture for a more efficient oil-water separation.

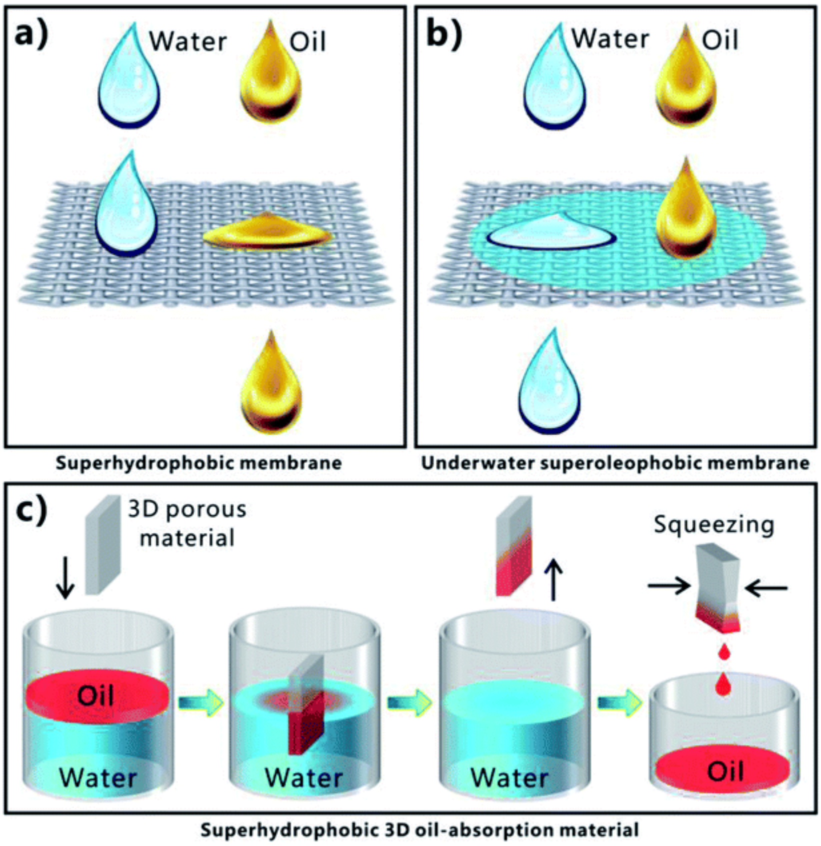


Figure 2. The oil-water separation process by using different superwetting porous materials with different separating ways. (a) superhydrophobic porous membrane. (b) pre-water-wetted underwater super-oleophobic porous membrane. (c) superhydrophobic/super-oleophilic 3D porous material (Song *et al.*, 2014).

1.7 Introduction of zein

Zein, a natural biopolymer mainly obtained from corn, is one kind of prolamins with molecular weight about 19-22 kDa （Girija et al., 2012）. Zein has more than 60% hydrophobic amino acids, which makes it amphiphilic and highly potential to form hydrophobic material (Shukla and Cheryan, 2001). It has advantages such as good biodegradability, biocompatibility, and low toxicity comparing to manufactured synthetic polymers. Zein is widely applied in agriculture, cosmetics, packaging, and pharmaceuticals (Paliwal and Palakurthi, 2014). It can also self-assemble into various structures, including spheres, sponges, films, and fibers, which make it an attractive biomaterial for controlled drug release and tissue engineering (Paliwal and Palakurthi, 2014). Solvents used for zein nanofibrous films electrospinning include acetic acid, aqueous methanol, ethanol, and isopropyl (Selling et al., 2007). Additionally, zein is not water-absorbing comparing to other natural biopolymers.

Our previous study used electrospinning to form superhydrophobic/hydrophobic zein nanofibers (Dong et al., 2013). It showed that the highest water contact angle of the surfaces could reach 155.5±1.4°. Further study about the formation mechanism of the electrospinning process showed that the hydrophobic groups of the zein amphiphiles tend to accumulate on the surface to make it superhydrophobic/hydrophobic. Zein based fibers have the potential of high oil absorption capacity and outstanding oil-water selectivity. However, there is little study about using zein for oil absorption and oil-water separation applications.

1.8 Spider silk

Spider silk, which enjoys a high reputation as a fiber with excellent mechanical properties, biocompatibility, and biodegradability. It has potential for processing in aqueous solution under ambient conditions make silk-based materials good candidates for technical textiles, and biomedical engineering such as drug delivery systems and scaffolds for tissue engineering (Altman, Diaz, Jakuba et al.,2003; Wang, Kim, Vunjak-Novakovic, & Kaplan., 2006; Frandsen, & Ghandehari, 2012; Vollrath & Proter, 2006;

Vendrely & Scheibel, 2007). Another intriguing ability is to collect water with the driving forces of surface energy gradient between the spindle-knots and the joints and the difference in Laplace pressure which is studied by Zheng et al. (2010). Naturally, water drop can stand on the spider silk network instead of being absorbed. It indicates that spider silk is hydrophobic. Silks are modular in design, with large internal consensus repeats, with a molecular weight of 200-350 kDa and above, flanked by shorter non-repetitive N- and C- terminal domains (~15 kDa). Spider silk is a natural block copolymer, where hydrophobic and hydrophilic regions in its molecule are linked to organizing into a functional material with exceptional properties. Spider silk has the potential of oil absorption, however, little study has been conducted.

Recently, researchers have worked to gain a better understanding of the molecular structure of spider silk and the mechanism of its formation naturally, in the purpose to synthesize an alternative for biomedical research and industrial applications. Among the different parts of spider silk net, draglines are most intensely studied. Dragline silks are utilized by orb-weaving spiders to build frame and radii of their nets and as lifelines that are permanently dragged behind. The dragline silk is composed of two similar proteins, the major ampullate spidroin 1 (MaSp1) and 2 (MaSp2) (Holland, Mou, & Yarger, 2013). In the repetitive region, tandem repeat units of glycine-rich sequences are followed by stretches of polyalanine sequences (Ayoub, Garb, & Desalle, 2007). They are found to form less ordered helices as well as β-sheets contributing to flexibility (Fu, Shao, & Fritz, 2009). β-turn structure, when repeated, forms spirals, similar to that of elastin (Xiao, Senbo, Xiao, Shijun, & Grter, Frauke. (2013). A large amount of βspiral structure gives elastic properties to the capture part of the orb web (Yang, Yang, Chen & Yao, 2008). The formation process of the silk fibers *in vivo* is a protein assembly process in the ampullate gland. The silk proteins assemble along with the elongational flow to liquid-solid phase transition when proteins are accumulated and stored at remarkable high concentration (50 w/v) (Jin, H., & Kaplan, D. 2003). The liquid crystalline fluid further aligns along the flow axis in a pre-aligned manner to form β-sheet crystals and fiber is released as the stress increases. The pulling force induces structural changes to the fiber, especially to the terminal regions, by exposing the hydrophobic regions allowing for phase separation between solvent (water) and the proteins (Eisoldt, Smith, & Scheibel. (2011). Further rearrangement of the core region within the micellar structures, in addition, the mechanical forces lead to phase separation and protein assembly.

Scientists found that the content of β-sheet has an impact on the hydrophobicity of spider silk. Drop cast of spider silk protein solutions on different hydrophobic as well as hydrophilic templates out of different starting solvents (hexafluoroisopropanol, formic acid and aqueous buffers) generated silk films varying in structure and surface properties. According to the research, the used templates had an impact on the content of β-sheet structures resulting in hydrophobicities (Wohlrab, Spies & Scheibel, 2012). John et al. (2002) designed a study of the conversion of soluble form of the major spider-silk protein, spidroin, directly extracted from the silk gland, to a β-sheet enriched state using circular dichroism (CD) spectroscopy. And the results showed in the conversion process, there is a refolding of the structure to enrich for β-sheet structure.

As the water-soluble spider silk protein transferring to hydrophobic spider silk, the hydrophobicity may be contributed by β-sheet structure. Wang et al. (2012) studied the mechanism of zein self-assembly at the nanoscale and found that in the evaporation-induced self-assembly process, the original α-helices of zein transfers to β-sheet structure, which is similar with the fabrication of spider silk.

1.9 Introduction of gelatin

Gelatin, a well-known bio-based protein, can be easily found in animal skin, bone and myolemma (Morsy et al., 2017). It has been extensively used in food, chemical engineering and pharmaceutical industries due to its low cost, biocompatibility and biodegradability. Gelatin has abundant hydroxyl, carboxyl and amino groups in its molecule chains, offering it with easy gelation and functionalization. Gelatin can be set as the starting material to build 3D structures. Porous gelatin networks have been developed for tissue engineering, flame retarding, oil/water separation and pollutants adsorption, separately.

Gelatin is one of the most commonly used biopolymers approved by the Food and Drug Administration of the United States, and it is rich in content and has good biocompatibility and biodegradability (Steyaert et al., 2016). Gelatin nanofibrous films for bioactive encapsulation and tissue engineering have been prepared by electrospinning with solvents like 2,2,2-trifluoroethanol, acetic acid, and formic acid (Steyaert et al., 2016; Fu et al., 2014; Báez et al.,2005; Okutan et al., 2014). Steyaert and colleagues found that the gelatin nanofiber membranes were soluble in cold water because of the high surface-to-volume ratio (Steyaert et al., 2016).

1.10 Electrospinning

Electrospinning is a process applying a strong electric field to fabricate fibrous polymer mats (Bhaedwaj et al., 2010). The laboratory equipment consists of a spinneret connected to a high-voltage power supply, a conductive capillary connected to a liquid storage tank containing polymers, and a collector. After exceeding a critical voltage, the repulsive forces overcome the surface tension of the solution and ejects charged jet from the spinneret. With the acceleration of charged jet to the low potential region, the solvent evaporates while the entanglements of the polymer chains prevent the jet from breaking up, thus forming ultrathin fiber (Doshi and Reneker, 1995). Electrospinning is a simple, versatile, and promising method for preparing continuous nanofiber membranes from polymers (Bhaedwaj et al., 2010). Because of the high surface-to-volume ratio and high pores density, these fibers are outstanding candidates for various applications, such as filtration, wound healing, scaffolds in tissue engineering, drug delivery, enzyme immobilization, biosensors and so on (Chen et al., 2017; Gupta et al., 2014; Zhang et al., 2017).

Chapter 2: Materials and Methods

2.1 Chemicals

Zein was purchased from Wako Pure Chemical Industries, Ltd. (Tokyo, Japan). Ethanol (96% v/v) was obtained from Guangdong Guanghua Sci-Tech Co. Ltd. (Guangzhou, China). 50% glutaraldehyde aqueous solution, 100% tert-butanol (TBA) and citrate acid (solid) were purchased from Shanghai Aladdin Bio-Chem Technology Co., Ltd. (Shanghai, China). Ethanol, toluene, acetic acid (99.8%), THF, acetone, 1, 2-dichlorobenzene, petroleum ether, hexane, and dichloromethane were purchased from Duksan Pure Chemicals Co. Ltd. (South Korea). Spider silk was obtained from Taobao. Gelatin (Type B, Bloom 250, MW, 100 kDa) were obtained from Sigma Aldrich (St. Louis, MO, USA).

2.2 Instruments

FTIR spectrum was obtained by Micolet iS50 FT-IR Spectrometer equipped with iTR ATR unit (Thermo Fisher Scientific Inc., USA). Thermogravimetric was measured by TGA/DSC (Netzsch STA 449C, Jupiter). SEM images were obtained using JEOL JSM-6490 SEM (Tokyo, Japan). Water contact angles were measured by a standard goniometer (Kruss GmbH DSA 100, Hamburg, Germany). The mechanical strength was tested by a universal tester (AJS-J. Shimadzu, Japan). The CD spectra were recorded using a JASCO J-810-450S spectropolarimeter.

2.3 Electrospinning and the formation of 3D nanofiber framework

25 wt% zein solution was prepared by dissolving zein in 80% (v/v) aqueous ethanol solution. A nanofiber electrospinning unit (Kato Tech Co., Ltd., Tokyo, Japan) was used. The negative electrode was linked to a drum collector covered with an aluminum foil, which was rotating at 1000 rpm. The positive electrode was connected to a metal needle with an inner diameter of 0.9 mm. The metal needle was connected to a 10-ml syringe containing zein solution. The syringe was placed horizontally on the controlled injection pump and keep the flow rate at 2 ml/h. The needle was horizontally directed towards the collector, and the distance from the tip to the collector was 15 cm. Because the set voltage difference between the needle and the collector is 20 kV, the positive charge of zein solution is sprayed and deposited on the collector. After 2 h of electrospinning, the samples (0.7 mm thick) were collected and removed from foil after overnight drying.

The obtained nanofiber mats were cut into small pieces and 1g of nanofiber was dispersed in 50, 75, 100, 125, 150, 175, 200, and 300mL TBA, respectively, and the samples were named E50, E75, E100, E125, E150, E175, E200, and E300 accordingly. Then the nanofiber suspensions were homogenized with IKA T18 homogenizer at 7500 rpm for 30 min. The nanofiber dispersions were poured into the 24-wells plate, with 2 mL mixture per well, and frozen at -20°C for 0.5 h, then, freeze-dried for 24 h. Each well of the plate was in a cylindrical shape, with a diameter of D =14mm and a height of h=10mm, which was in the same size as each of the scaffolds. The porosity of the scaffolds was calculated using the following formula:

$$\emptyset = 100 * (1 - \frac{\rho_a}{\rho_c})$$

where ρ_a was the apparent density of the scaffold and ρ_c was the density of zein fiber (1.25 g/cm^3) (Cook, 1984) and for spider silk ρ_c was 1.34 g/cm^3 (Saravanan, 2006). The apparent density (ρ_a) of the scaffolds was calculated using $\rho_a = M/V_a$, where M is the mass of the scaffold and V_a is its apparent volume, which is measured by ruler. So, $\rho_a = \frac{M}{[h\pi\left(\frac{D}{2}\right)^2]} = \frac{4M}{\pi h D^2}$.

Various methods were applied to make scaffolds cross-linking to enhance the mechanical strength. Non-crosslinked scaffold (NS) were placed into an oven under constant heating at 150°C for 30, 60, and 90 min, respectively, and obtained the heat-treated scaffolds (HS). NS were immersed into the citrate acid - TBA mixtures with citrate acid concentrations of 1%, 3%, 5% and 10% (w/w), respectively, for 24h and obtained the citrate acid cross-linked scaffolds (CS). NS were placed in the saturated vapor environments produced by 99%, 80%, 50%, and 0% ethanol-in-water solutions, respectively, and obtained the solvent vapor induced cross-linked scaffolds (SVS). The appearances of scaffolds were observed and recorded in every hour. NS were immersed into the glutaraldehyde (GA) - TBA mixtures with the concentrations of 0.5%, 1.5%, 2.5%, and 5% (w/w), respectively, for 24h and obtained the GA solution cross-linked scaffolds (GASS). NS were placed in the saturated vapor environments produced by 50% and 25% glutaraldehyde aqueous solutions for 24h, 48h, and 72h, respectively, and obtained the GA vapor cross-linked scaffolds (GAVS).

2.4 Preparation of zein/gelatin nanofibers

Zein protein solutions (30% (w/v)) was prepared by dissolving zein in solvent, either acetic acid/water (8:2) mixture (AA) or acetic acid/ethanol/water (6:2:2) mixture (AE). Blends of gelatin and zein at weight ratios of 0/1, 1/3, 1/2, 1/1, 2/1, 3/1, and 1/0, were named as GZ01, GZ13, GZ12, GZ11, GZ21, GZ31, and GZ10. Nanofibers were formed using an electrospinning unit (Kato Tech Co., Ltd., Tokyo, Japan). The negative electrode was linked to a drum collector covered with an aluminum foil, which was rotating at 1000 rpm. The positive electrode was connected to a metal needle with an inner diameter of 0.9 mm connected to a 10 ml syringe containing zein solution. Place the syringe horizontally on a controlled syringe pump and kept the flow rate at 1 ml/h. The needle was horizontally directed towards the collector, with the 10 cm distance from the lip to collector. Due to the set voltage difference of 15 kV between the needle and the collector, the positive charge of zein solution is sprayed onto the collector. After electrospinning for 2 h and drying overnight, the nanofiber samples (0.7 mm thick) were collected.

2.5 Scanning electron microscopic studies

The morphology of nanofibers was observed using a JEOL JSM-6490 scanning electron microscopy (SEM, Tokyo, Japan). The samples were gold coated (300 Å) with an Edwards S150B sputter coater to help improve electrical conductivity and observed under the SEM. The fiber diameters and their diameter distributions were determined by Nano Measure software by averaging 40 fibers from each of the 5 randomly selected images.

2.6 Porosities

The nanofibers were cut into pieces with equal size (1 x 1 cm^2) and weighted using a high accuracy balance (AB104-S, Mettler Toledo, Italy). The thickness was measured using a digital micrometer by averaging three random selected spots. The apparent density of nanofibers (ρ_A) was calculated based on the weight and volume of the nanofibers. And the material density (ρ_M) was calculated based on the density of gelatin (1.41 g/cm^3) and zein (1.22 g/cm^3) and their mass percentages. The porosity was determined using the following equation:

$$\%\text{porosity} = \left[1 - \left(\frac{\rho_A}{\rho_M}\right)\right] \times 100$$

where ρ_A is the apparent density of the nanofibers, and ρ_M is the material (composite gelatin and zein) density.

2.7 Water contact angles

The hydrophobicity of nanofibers was assessed using the WCAs. It was measured using a standard goniometer (Kruss GmbH DSA 100, Hamburg, Germany). During the test, 5μl water drops were introduced using a micro-syringe system and dispensed drop by drop on the surface of the samples, and images were captured using a camera system. The WCAs were fit according to the Young–Laplace equation by the software. For each sample, different positions on the surface were selected for repeated measurements to obtain an average.

2.8 Fourier transform infrared spectroscopy

The ATR-FTIR spectra of nanofibers were recorded using the Nicolet IS50 spectrometer in mid-IR mode. It equipped with a Universal ATR (Attenuated Total Reflectance) sampling device containing diamond/ZnSe crystal. Spectra were acquired and processed with the Spectrum software version 6.3.2. The spectra were scanned at room temperature in transmission mode over the wave number range of 4000-650 cm^{-1}, with a scan speed of 0.20 cm/s. 30 accumulations at a resolution of 4 cm^{-1} were used. Triplicates of each sample were averaged to obtain an average spectrum. A background spectrum of air was scanned under the same instrumental conditions before each series of measurements. Before the calculation of the secondary derivatives, the same processing of spectra smoothing was applied to each spectrum to improve the signal-to-noise ratio. The Savitsky−Golay derivative program was used to calculate the secondary derivative.

2.9 Thermogravimetric analysis

To study the thermal decomposition and phase transition of the nanofibers, thermogravimetric analysis (TGA) was conducted using Mettler Toledo TGA/DSC3+ under a nitrogen atmosphere (20 ml/min). Sample were sealed in 70 µl aluminum pans and heated from 50 to 500 °C with a heating rate of 10 °C/min.

2.10 Mechanical properties

The mechanical strength was tested by a tensile test. Fiber samples were cut into 40 x 10 mm strips. The mechanical properties were measured by a universal tester (AJS-J, Shimadzu, Japan) equipped with a 5 N capacity load cell. Data of modulus of elasticity, tensile strength and elongation at break were reported from the stress-strain curves with an average value ± SD (n = 5).

2.11 Oil absorption ability

To examine the absorption capacity of nanofibers, various oils and organic solvents, including olive oil, motor oil (50w), diesel oil, petroleum ether, hexane and chloroform were used. Nanofibrous membrane with the size of 40 mm (length) x 10 mm (width) was first weighed at room temperature, and then immersed in 100 ml oil or solvent for 0.5h to reach equilibrium. Then, the nanofibrous membrane was removed and drained in air for 30s and weighed again. The absorption capacities (k) of the nanofibers are calculated using the following formula:

$$k = (m_{saturated} - m_{initial}) / m_{initial}$$

where $m_{saturated}$ is the weight of the nanofibers after absorption and $m_{initial}$ is the weight before absorption.

2.12 Selective absorption and oil-in-water emulsion removal test

The toluene and deionized water (1:9, v/v) was mixed under ultrasonic condition to obtain the oil-in-water emulsion. The nanofibrous membrane was forced into the emulsion to remove oil droplets. After oscillating for a certain time, the nanofibrous membrane was taken out.

2.13 Multifunctional Oil-Water separation

To obtain a heavy metal contaminated oil-water mixture, 40 ml of hexane was mixed with 40 ml of a Cu^{2+} aqueous solution. Then, the nanofibrous membrane was added into the mixture for 24h with stirring. After that, the membrane was collected and EDS was performed to check the adsorption of Cu^{2+}.

2.14 Circular dichroism spectroscopy

Zein with different concentrations (0.0125, 0.025, 0.05, 0.1, 0.2, 0.4, 0.8, 1.6, 3.2 mg/ml) was dissolved in 10 mL ethanol-water mixtures with EtOH % = 85, 75, 65, 55, 45, 35%. Then the solution was sonicated with a VC-750 ultrasound generator (Sonics and Materials, Inc., Newton, CT) at 300 W for 5 min. Sample solutions was collected and the CD spectra were obtained.

The CD spectra were recorded using a JASCO J-810-450S spectropolarimeter. Experimental conditions used were as follows: bandwidth = 2 nm; scan speed = 50 nm/min; resolution = 1 nm; and accumulation with 1–10 traces for averaging. Experiments were performed in a quartz cell with a 0.2 cm path length over the range of 190–250 nm at 25 °C. The content of α-helix, β-sheet, and random coil was calculated by the software.

Chapter 3: Sustainable and Biodegradable Bioinspired Superhydrophobic Nanofibrous Scaffold for Oil/water Separation

3.1 Introduction

Oil spills and chemical water pollution accidents lead to enormous loss of valuable energy resources and generate long-term influences on the water environment and marine ecological systems. Therefore, scientists have made great efforts to look for effective cleaning methods and materials that can clean up the oil, and the reduction of ecological pollution is in high demand (Havase, Kanamori, Fukuchi, Kaji, & Nakanishi, 2013). A variety of techniques, including physical absorption by absorbent materials, physical diffusion, mechanical cleaning by oil skimmers, burning, and biodegradation, have been employed to purge oil spills. Among these methods, physical absorption using absorbent materials is generally considered to be one of the most efficient countermeasures owing to their special properties of simple, efficient, and fast, (Zhou, Zhang Xu, Men, & Zhu, 2013; Peng, Yuan, Yan, Yu, & Luo, 2014). It also has many other advantages, including less energy requirement and good recovery ability. However, the poor water-oil separation ability of sponges limited their practical oil absorption capacity, because they could absorb water and oil at the same time. Consequently, the ability of selectively absorbing oils and completely repelling water is necessary, which required excellent hydrophobicity.

In this study, the hydrophobicity and oil absorption capacity of spider silk and zein fiber were investigated. The molecular structure of zein fiber and spider silk was characterized and compared by ATR-FTIR. The α-helix to β-sheet transformation of zein was explored by CD spectroscopy. To further study the factors of oil absorption capacity, porous zein-based nanofibrous scaffold was prepared using a two-step method. The method is simple, low-cost, with short fabrication duration. The hydrophobic scaffold could absorb various oils and organic solvents and hold its structure in aqueous condition. The morphology was characterized using SEM and the hydrophobicity was characterized by water contact angel. The oil-absorption capacity and selectivity were also tested. Effects of different porosity, different ratios of the disperse solvents and fibers, the density and viscosity of oils on the effects of oil absorption using this scaffold were also investigated.

3.2 Results and discussion

3.2.1 Apparent density and porosity

Table 5. Chemical compositions and physical properties of various frameworks.

Sample	Mass of fiber (g)	Diameter (mm)	Height (mm)	Apparent Density ρ_A (g/cm^3)	Porosity (%)
E150	0.0126	14	8	0.0108±0.0011	99.1
E125	0.0153	14	8	0.0141±0.0053	98.9
E100	0.0191	12	8	0.0211±0.0010	98.3
E75	0.0162	16	9	0.0090±0.0005	99.3
E0	0.0084	1.2	0.15	0.495±0.279	60.4
SSFS	0.0014	0.3	0.23	0.861±0.486	35.7

Nanofibrous scaffolds were obtained after homogenization and freeze-dried. However, when the ratio of fiber and TBA was higher than 1:50 (E50), it was difficult to homogenize and form scaffold. On the other hand, when the ratio was lower than 1:200 (E200& E300), it was hard to form 3D structure because of the low concentration of zein fiber. As Table 5 shown, the density increased as the ratio increased. E0 has the highest density of 0.495 g/cm^3. In the process of sample preparation, volume is shrank after freeze-dried. The abnormal lower apparent density of E75 may be caused by the intensive viscosity and inhomogeneous of the dispersion mixture, due to relatively higher fiber concentration. Therefore, the shrinkage of E75 is the lowest one among all samples so the volume is higher and density is lower. According to Table 5, porosity is in the following order: E75 > E150 > E125 > E100 > E0 > SSFS. In general, the porosity decreased as the ratio increase. The abnormal higher porosity of E75 may be caused by its high apparent density.

Table 6. Performance of crosslinking agent.

Scaffold	**Observation**	**Product**
NS	-	-
HS	Shrinkage, browning	Small, yellow, fragile scaffold
CS	Not reaction	-
SVS	Intensive shrinkage	Zein ethanol solution
GASS	Scaffold dissolved	-
GAVS	Shrinkage	Slightly shrinking scaffold, tight membrane on surface

There is no perfect cross-linking agent for zein scaffold. GA vapor seems to be the best choice, but the hidden toxicity of GA residue should draw our attention.

3.2.2 Morphologies of the hydrophobic scaffolds

Figure 3 showed the SEM images of the nanofibrous scaffolds and SSFS. The scaffold has a dimension of 14 mm (diameter) x 10 mm (thickness) and same shape of its reaction container. As the SEM images shown, the nanofibrous scaffolds are formed via the physical attachment of zein fibers. In this work, TBA can diffuse zein fibers and increase the porosity of scaffolds. In contrast to the mesopores (2-50 nm) of the scaffolds formed by the zein nanofibers, highly porous structures of the scaffolds with macropores (>50 nm) can be clearly observed in SEM images of Fig. A-F. Their macropores are possibly caused by the volume of diffusing reagent which was removed by freeze-dried. However, an increase of volume of TBA from 75 to 150 ml does not significantly impact the scaffold structures as shown in Figure 3. Comparing (A-D) with E, the fiber was cut into pieces in the process of homogenization. The SEM images show that the morphology of E0 is similar to that of SSFS that both are smooth fibers.

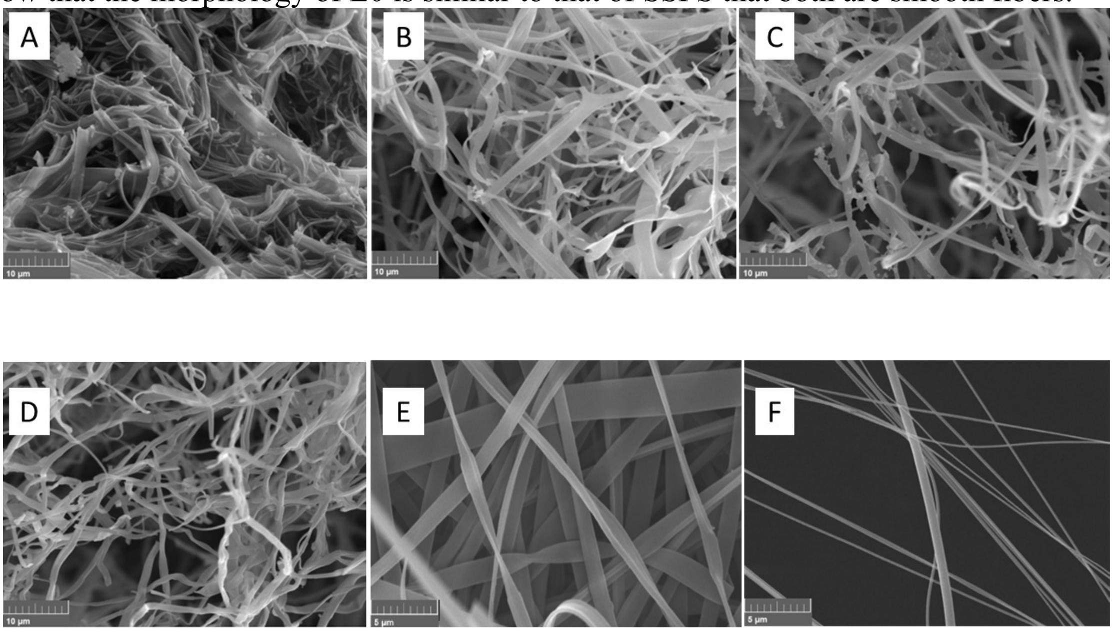


Figure 3. SEM images of non-crosslinked scaffold: (A) E75, (B) E100, (C) E125, (D) E150, (E) E0 and (F) SSFS. (A-D): Scale bar = 10μm, (E-F): Scale bar = 5μm.

3.2.3 Water contact angle (WCA)

In order to investigate the hydrophobicity of the developed nanofibrous scaffolds, the WCAs were measured. As shown in Figure 5, the WCAs of E75, E100, E125, E150, E0, and SSFS were 106.120°, 100.233°, 103.125°, 98.265°, 110.743°, and 106.674°, respectively. It indicated that the surfaces of the zein scaffolds and SSFS are hydrophobic. The WCAs of scaffolds are around 100-110° except E150, hence the reagent ratio has limited influence on hydrophobicity. It is well-known that the contact angle of water intensely depended on the functional groups on the surfaces. Therefore, in this case, such a small change may be the result of the same surface structure. For the zein casting film, the WCA was 42.864°, so the surface is hydrophilic. Similarly, for spider silk solution casting film, the water contact angle was 0°, which is also hydrophilic. According to the result, zein based nanofibrous scaffolds and SSFS are hydrophobic while zein casting film and spider silk casting film are hydrophilic. Therefore, electrospinning made the zein surface changed surface.

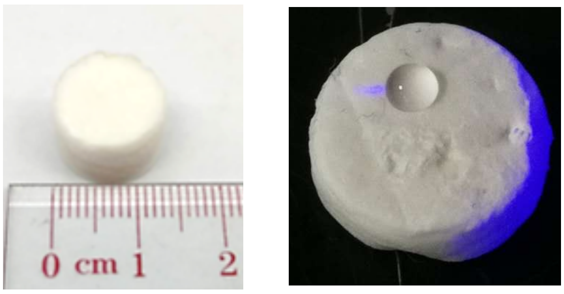


Figure 4. The photographs of scaffolds.

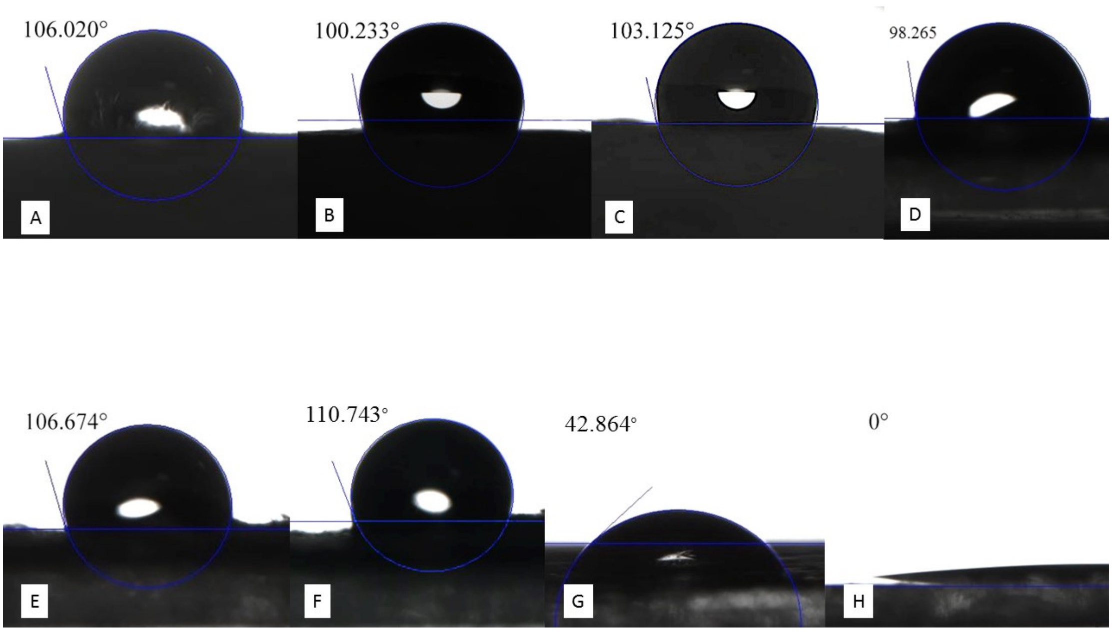


Figure 5. Water contact angles (A)E75, (B)E100, (C)E125, (D)E150, (E) SSFS, (F)

E0, (G) zein solution casting film (2.5g/ml 75% ethanol), and (H) spider silk solution casting film.

3.2.4 ATR-FTIR

To investigate the structure difference or changes, FTIR spectra of zein based nanofibrous scaffolds, and SSFS, respectively, are obtained and showed in Figure 6 and 7. Spectra were recorded in the 4000-850 cm^{-1} frequency region. In both Figure 6 and 7, a large band at 3400-3200 cm^{-1} is mainly related to the hydroxyl groups of O-H stretching vibrations (Knill, & Kennedy, 2000). In Figure 6 the most sensitive spectral region to protein secondary structural components is the amide I band (1700 to 1600 cm^{-1}), which is almost entirely due to the C=O stretch vibrations. The amide II band (1600 to 1500 cm^{-1}), in contrast, is derived mainly from N-H bending and from the C-N stretching vibrations (Li, Lim, & Kakuda, 2009). In Figure 6, fingerprints at 1647 cm^{-1} (amide I band), 1538 cm^{-1} (amide II band) and 1450 cm^{-1} (methylene), close to the character peaks of E0 at 1641 cm^{-1}(amide I band), 1530 cm^{-1}(amide II band) and 1452 cm^{-1} (methylene) were shown in the spectra of E75, E100, E125, and E150. Therefore, the formation process of scaffold has limited influence on the molecular structure. According to the spectra in Figure 6, the fingerprints of spider silk at 1632 cm^{-1}(amide I band), 1522 cm^{-1}(amide II band), 1452 cm^{-1} (methylene). Therefore, the chemical structure of SSFS is similar with E0.

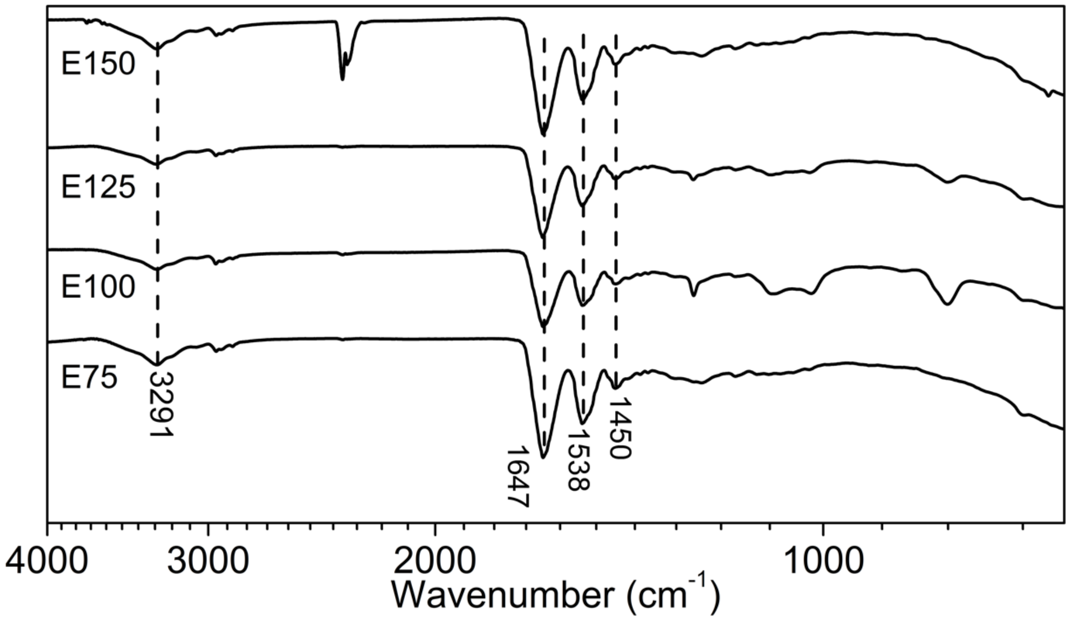


Figure 6. FTIR spectra of scaffolds E75, E100, E125, and E150.

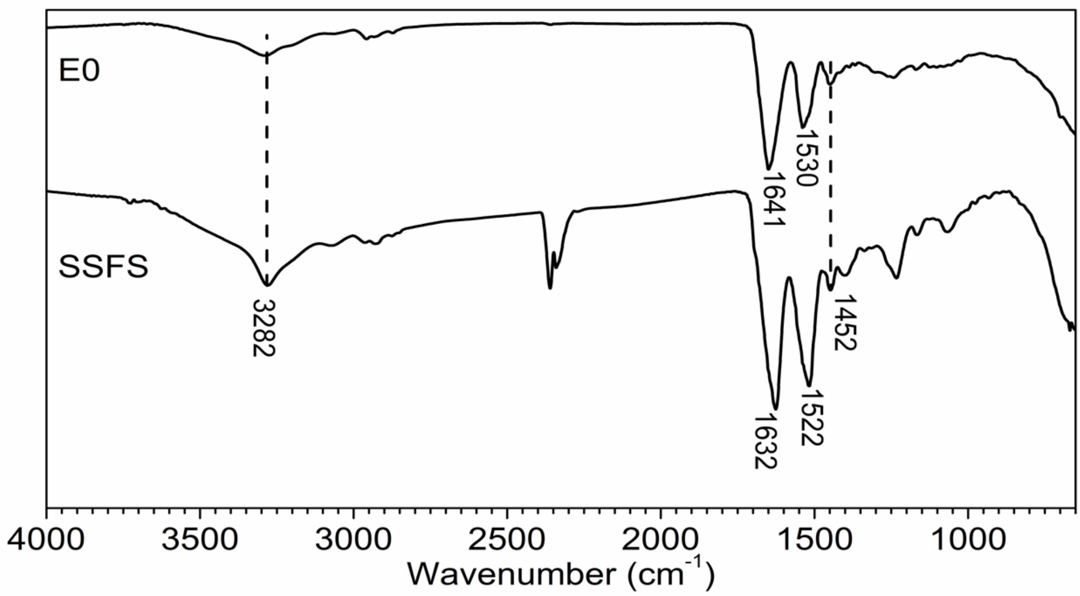


Figure 7. FTIR spectra of E0 and SSFS, respectively.

3.2.5 Oil absorption ability

Maximum oil absorption capacity

The above results show that scaffold has low density and high porosities. This highly porous structure should give it the ability to absorb organic liquids. The hydrophobicity of scaffold indicates their potential of selectively absorb oil in oil-water mixture. To confirm the hypothesis and study the absorption process, a scaffold was put into oil-water mixture and optical images of the oil absorption process were obtained. The scaffold could quickly absorb oil which was attached to its surface, and the scaffold was slightly moved to other parts after absorbing all the oil attached. When a nanofibrous scaffold was placed in a motor oil (dyed with Coumarin 6)-water mixture (Figure 8), the scaffold quickly and selectively absorbed the motor oil from the mixture and left a clear ring around the scaffold. It usually only took a few minutes to complete the absorption and the oil filled scaffold floated on the water, which facilitated its collection. The oil filled scaffold is sticky but it maintained its original shape. When the scaffold was pressed, some oil trickled down the sides. However, the most of oil was locked in the scaffold and was not released without external force. Therefore, the scaffolds could be used to separate oil from water.

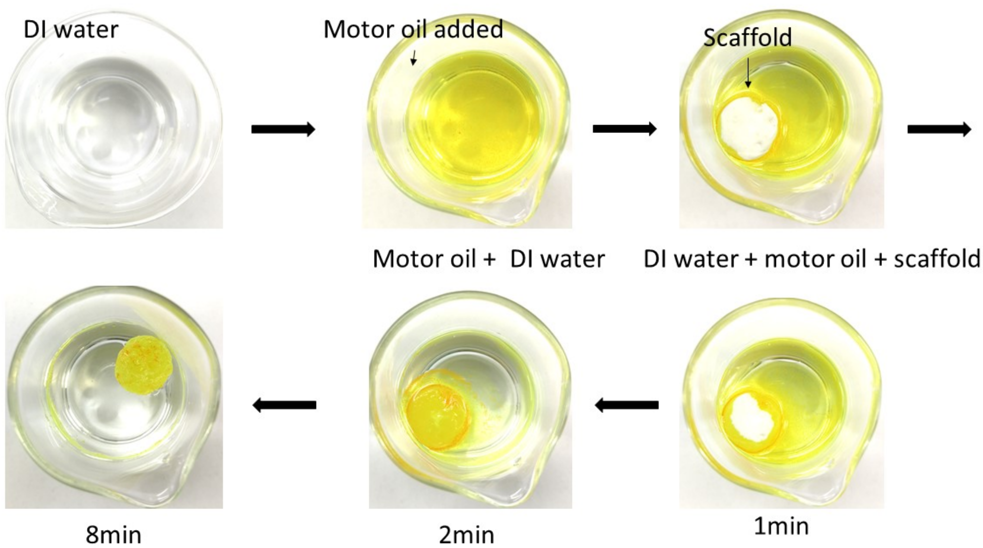


Figure 8. Digital images of scaffolds immersed in motor oil-water mixture (E150).

Figure 9 shows the absorption capacities (k) of E75, E100, E125, E150, E0 and SSFS for different oils (motor oil, diesel oil, olive oil and pump oil) and various organic solvents (toluene, THF, acetone, 1, 2-dichlorobenzene, petroleum ether, hexane, and dichloromethane). The k values for all the set of scaffolds and SSFS were high, especially for E0, which had k values around 150g/g for diesel oil and pump oil. The E0 has a maximum k value of 152 g/g, which is an extraordinary high value. Graphene sponges reported by Be et al. had a maximum k of 87 g/g (Bi, Xie, Yin & Ket al., 2012), whereas CNF aerogels had a k value of 110 g/g (Liang, Guan, Chen, Zhu, Zhang & Yu, 2012), and the RGO form had a maximum k of 122 g/g (He, Liu, Yue, Wu & Tao et al., 2013). The absorption ability of all carbonaceous absorbents is much higher than that of organic absorbents, for example, polydimethylsiloxane sponges with k value only 11 g/g (Choi & Kwon, 2011). The comparison of the absorption of oils by various materials are listed in Table 7. The scaffolds and spider silk have polar functional groups and are hydrophobic, so they show a stronger affinity for oil which results in a higher absorption capacity. This phenomenon is similar to polymeric absorbents. Although E150, E125, E100, and E75 had higher porosity than that of E0, their k values were lower than that of E0. It may because that the main fact of oil absorption is hydrophobicity, and porosity has limit influence on k value. On the other hand, the air of scaffold occupied the porous structure is not excluded completely. Air can become sealed in the pores and it occupies some of the volume. Besides, from the SEM images in Figure 3 the fiber in E0 is a complete line while others were cut into pieces by shear force, and oil is absorbed and transferred along the fiber, which may have a positive impact on oil absorption.

Therefore, the absorption capacity is comparatively low. SSFS also had good oil absorption capacities (motor oil: 53 g/g; diesel oil: 29 g/g; olive oil: 52 g/g; pump oil: 32 g/g).

Table 7: Comparison of the absorption of oils by various materials.

Sorbents	Absorption capacity (g/g)	Ref.
Polyurethane foam	80	(Jeong, Park & Lim, 2016)
Butyl rubber	23	(Ceylan, Dogu, Oya & Okay et al., 2009).
Polypropylene	15.7	(Ceylan, Dogu, Oya & Okay et al., 2009).
Cotton fiber	20	(Deschamps, Caruel, Bonnin & Vignoles, 2003).
Spongy Graphene	87	(Bi, Xie, Yin & Zhou, et al., 2012).
Exfoliated graphite	75	(Zheng, Wang & Inagaki, 2004)
Sepiolite	0.18	(Rajaković-Ognjanović, Aleksić, & Rajaković. 2008)

The scaffolds also showed quick absorption of organic solvents. For zein based scaffolds, E150 produced the best organic solvent absorption capacities. The organic solvent capacity value of E150 was 95 g/g for dichlorobenzene, and this was the most highly absorbed organic solvent. The solvent uptake capacities were also studied in toluene, THF, acetone, 1,2-Dichlorobenzene, pertroleum ether, and hexane, the value was found 60, 41, 51, 78, 45, and 47 g/g respectively. These results indicate that the scaffolds could be used for absorbing organic solvents and oils from our environment and water surfaces with a high capacity. This could contribute to the cleaning of water surfaces and the environment of pollution caused by previously described events. Except THF, the organic uptake capacities of scaffolds were almost all in the order 1,2-Dichlorobenzene > DCM > toluene > Acetone > Hexane > Pertroleum ether, which is according to their density (Listed in Table 8). For THF, the scaffolds were shrunk so the absorption capacities were relatively low. SSFS also has excellent organic uptake capacities, especial for 1,2-Dichlorobenzene (96 g/g).

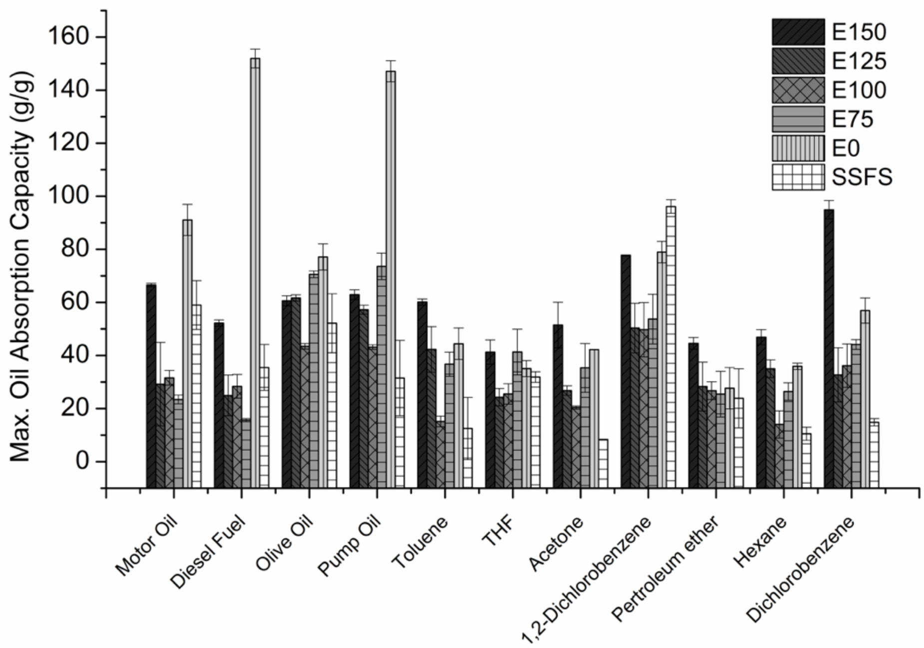


Figure 9. Maximum absorption capacities of various oils, including motor oil, diesel oil, olive oil, and pump oil, and organic solvents, including toluene, THF, acetone, 1,

2-dichlorobenzene, petroleum ether, hexane, and dichloromethane, onto the scaffolds of E150, E125, E100, E75, E0 and SSFS, respectively, at 25°C.

Table 8: Density of organic solvents, including toluene, THF, acetone, 1,2-dichlorobenzene, petroleum ether, hexane, and dichloromethane.

Organic solvents	Density (g/cm^3)
toluene	0.867
THF	0.889
acetone	0.784
1, 2-dichlorobenzene	1.3
petroleum ether	0.653
hexane	0.655
dichloromethane	1.3

3.2.6 Reusability

Figure 10 shows the varying values of absorption capacities of the prepared scaffolds for olive oil when the scaffolds are subjected to successive absorption–squeezing cycles. For all these samples, the oil absorption capacity is decreased as the cycle number increased. E150, E125, E100 and E75 were all broken after a few absorption-squeezing cycles due to their poor mechanical ability and weak cross-linking. Among E150, E125, E100 and E75, the reusability of E150, E125 and E100 is much better than that of E75, and the cycle number is 11, 17, 11 and 2, respectively.

E125 had the highest cycle number while E75 was broken after 2 cycles. Oil absorption capacities of E125 reduced gradually from 61 to 22 g/g after 17 cycles, with about 36% oil sorption capacity remained. E0 and SSFS were not broken in absorption-squeezing cycles and the cycle number was as high as 30, but the oil absorption capacity was reduced gradually with around 50% remained after 30 cycles. E0 has the highest oil absorption capacity at each cycle, which is in consistent with the result of maximum oil absorption capacity. The absorption and desorption processes of different oils are shown in Figure 11, and the recovery percentage by compression is viscosity-dependent that after 10 cycles, diesel oil with the lowest viscosity shows 68.5% recovery, olive oil shows 66.1% recovery, motor oil shows 48.1% recovery, while the most viscous pump oil exhibits a recovery of 37.5%. Figure 10 also shows multiple absorption−desorption cycles for diesel oil, where 65% of the saturated sorption capacity can be maintained over many cycles. Surprisingly, the recovery ratio is stable after the second cycle for all the four types of oil. Because of the interaction between oils and the scaffolds, part of the volume has been occupied by oil. Oil cannot be regenerated by mechanical extrusion, but the remaining pores can be fully utilized in recycling applications. Therefore, the stable recovery of absorbed oils is shown after the first cycle.

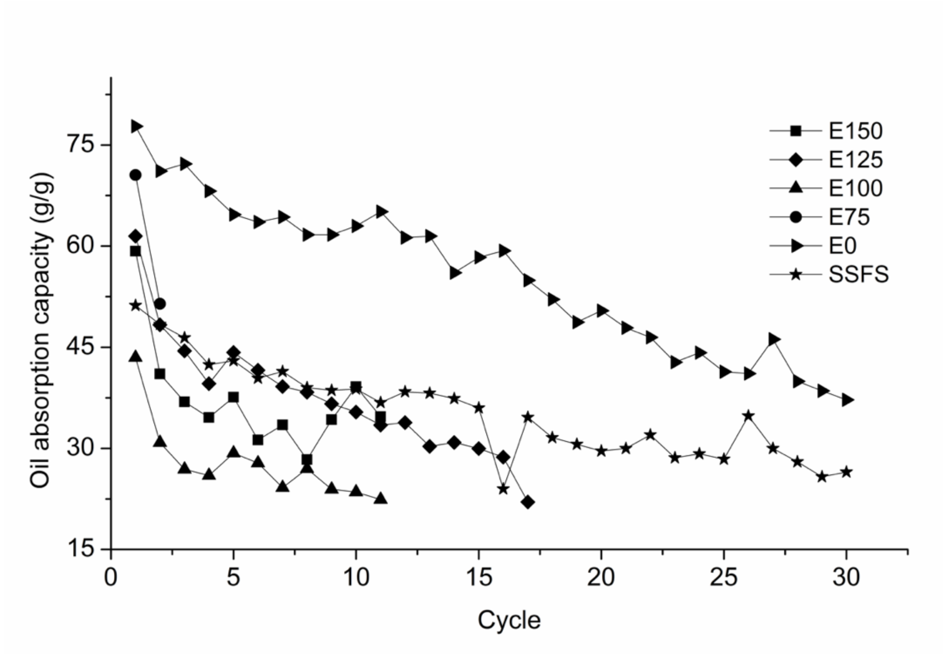


Figure 10. Absorption capacities of olive oil with various scaffolds, E150, E125, E100, E75, E0 and SSFS respectively, under different cycles.

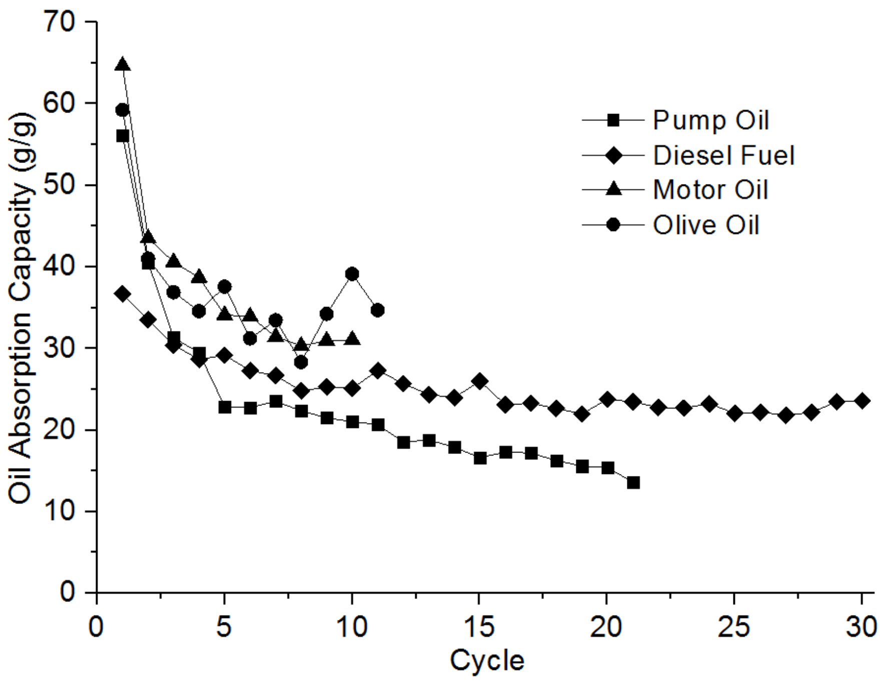


Figure 11. Absorption capacities of various oils with E150 under different cycles.

3.2.7 Circular Dichroism Spectroscopy

CD spectroscopy was used to characterize the molecular structure of zein solution and its conversion in the process of solvent evaporation. Samples of zein solutions with concentration 0.0125, 0.025, 0.05, 0.1, 0.2, 0.4, 0.8, 1.6, 3.2 mg/ml, respectively in different ethanol-water mixture with EtOH % = 85, 75, 65, 55, 45, 35%, respectively, were collected. The different ethanol concentrations was used to simulate the solvent evaporation process where the ethanol concentration is decreased as time increased since ethanol evaporates faster than water. CD spectra, collected at 25 °C in the 190–260 nm range, and the corresponding plot of β-sheet is shown in Figure 12. It shows the change in relative content of β-sheet during solvent evaporation. The content of β-sheet in all zein solution with different concentrations increased from an initial around 5% to 40% as the ethanol concentration decrease. Increasing the polarity of solvents can increase the content of β-sheet. The electrospinning of zein is a droplet-mediated or jet mediated evaporation-induced self-assembly process. Different from the evaporation method of zein film prepared by cast drying method, the zein solution is dried in the air in the shapes of droplets or jets. As discussed in the part of WCAs, electrospinning changes the hydrophobicity of zein solution. It may induced by the transformation of protein secondary structure, the increase of β-sheet content. Similarly, in the spider silk formation process, there is a refolding of the structure to enrich for β-sheet structure and the hydrophobicity is also increased.

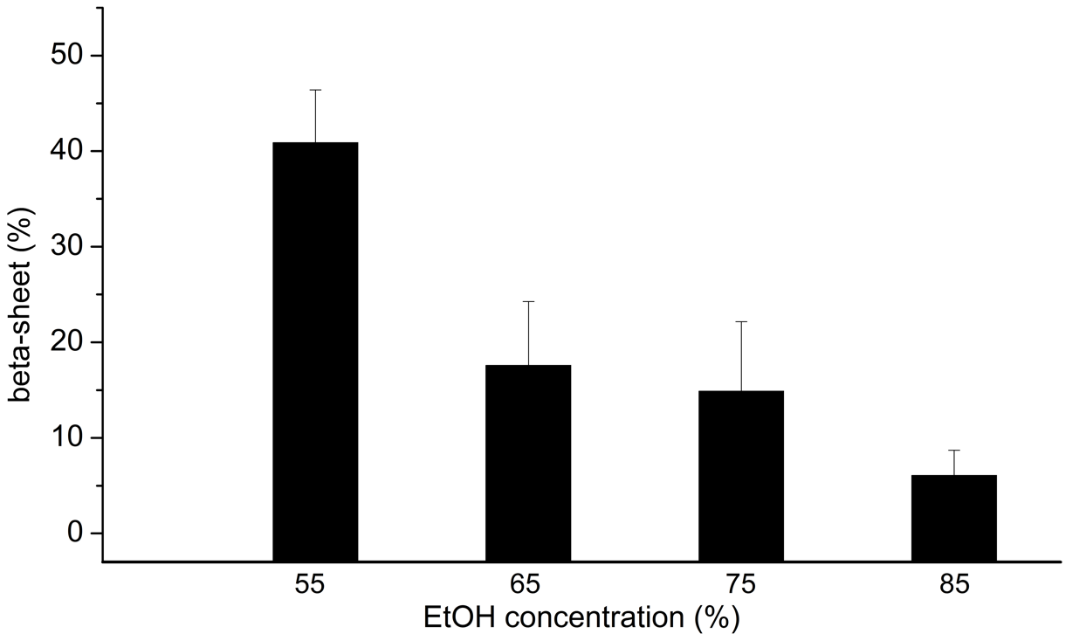


Figure 12. Circular dichroism spectroscopy of zein solutions (0.05, 0.1, 0.2, 0.4, 0.8, 1.6, 3.2 mg/ml) in different ethanol-water mixture (EtOH % = 85, 75, 65, 55).

3.3 Conclusion

Zein nanofibrous scaffolds were fabricated using electrospinning and the hydrophobicity and oil absorption capacity of scaffolds and SSFS were investigated. The surfaces of scaffolds (E75, E100, E125, E150 and E0) and SSFS were hydrophobic, with WCA of 106.120°, 100.233°, 103.125°, 98.265°, 110.743° and 106.674° respectively. The scaffolds showed extremely high absorbing abilities for organic liquids. The absorption capacities of the E0 in motor oil, diesel oil, pump oil, and olive oil were 91 g/g, 152 g/g, 77 g/g and 147 g/g respectively. The oil absorption capacities of the SSFS in motor oil, diesel oil, pump oil, and olive oil were 53 g/g, 29 g/g, 52 g/g and 31 g/g respectively. It was lower than that of E0, but higher than that of most oil absorbents. Porosity has limit influence on oil absorption capacities and the organic solvent absorption capacity is related to the density of solvent. E0 and SSFS has good reusability for 30 cycles while other scaffolds are broken in absorption–squeezing cycles. It is believed that the formation of hydrophobic surface was mainly contributed by the enriched β-sheet content. Zein fiber has similar molecular structure, hydrophobicity and oil absorption capacity with spider silk.

Chapter 4: Biodegradable Zein/Gelatin Composite Nanofibers for Oil/water Separation

4.1 Introduction

Oil pollutions caused by oil spills and industrial oily wastewater has now become serious environmental and ecological problems and greatly affected the quality of human life. Enormous efforts and various methods, including physical absorption by absorbents (Zhao et al., 2012), mechanical cleaning (Calcagnile et al., 2012), *in situ* combustion （Aurell and Gullett, 2010） and biodegradation (Boopathy et al., 2012), have been applied to address this issue. Among these methods, physical absorption using absorbent materials, such as activated carbon and synthetic polymers, is generally considered to be one of the most efficient methods, because of its efficiency, recoverability, protection of environment, less energy requirement, and good recovery ability (Zhang et al., 2012). Many strategies for preparation of absorbent materials with special wettability have been developed for the treatments of oil spills and oily wastewater, among which porous materials with hydrophobicity and oleophilicity are considered to be the best candidates (Adebajo et al., 2003).

Electrospinning is a process applying a strong electric field to fabricate fibrous polymer mats (Bhardwaj and Kundu, 2010). Electrospinning is a simple, versatile, and promising process to produce continuous nanofibrous films from polymers (Bhardwaj and Kundu, 2010). Due to the high surface area ratio and the density of pores in sub-micrometer length scale of the resulting fiber mat, these fibers are excellent candidates for various applications, including filtration, wound healing, tissue engineering scaffolds, drug delivery, enzyme immobilization, and biosensors (Chen et al., 2017; Gupta et al., 2014; Zhang et al., 2017). Zein nanofibrous films have been electrospun from acetic acid, aqueous methanol, ethanol, and isopropyl (Selling et al., 2007). Additionally, different from other natural biopolymers, zein is not water-absorbing. Zein based fibers have the potential of high oil absorption capacity and outstanding oil-water selectivity. There is no study about using zein for oil absorption and oil-water separation applications. Gelatin is one of the most commonly used FDA approved biopolymers, and it is abundant and has good biocompatibility and biodegradability (Steyaert et al., 2016). Gelatin nanofibrous films used for bioactive encapsulation and tissue engineering have been formed by electrospinning using various solvents, including 2,2,2-trifluoroethanol, acetic acid, and formic acid (Steyaert et al., 2016; Fu et al., 2014; Báez et al., 2005; Okutan et al., 2014). Steyaert and coworkers found that the gelatin nanofibrous film was cold-water-soluble due to the high surface-to-volume ratio (Steyaert et al., 2016).

In this study, gelatin/zein nanofibers were fabricated by hybrid electrospinning. The effects of gelatin/zein ratio and solvent on the morphology, diameter distribution, and porosity of the nanofibers were investigated. The fabricated nanofibers were characterized using SEM, ATR-FTIR, mechanical test and water contact angle measurements. The oil-absorption capacity was tested. Effects of the composition on the absorption capacity of various oils with different density and viscosity were also investigated.

4.2 Results and discussion

4.2.1 Morphologies and hydrophobicities of gelatin/zein fibers

Morphologies and hydrophobicities of gelatin/zein fibers fabricated using AA solution

The morphology of the electrospun gelatin/zein nanofibers using AA solution was studied using SEM and the results were shown in Figure 13. Generally, the fibers were randomly oriented and interlocking, which gave a highly porous structure. GZ01 (Figure 13A), when the fiber was made by 100% zein, spindle-like nanofibers with beads were shown and the average diameter of the fibers was 188.6 nm. After adding 25% gelatin to the zein solution (GZ13), round nanofibers with beads were produced (Figure 13B). As the content of gelatin was further increased to 33.3% and higher (GZ12, GZ11, GZ21, GZ31, and GZ10), the nanofibers were uniformly smooth without beads. The diameter of nanofibers was increased from 157.7 nm to 1282.7 nm as the content of gelatin increased from 33.3% (GZ12) to 75% (GZ31), but a further addition of gelatin to 100% (GZ10) decreased the diameter to 533.5 nm. The SEM results suggested that, by increasing the content of gelatin, the beads were disappeared and the nanofibers become smooth. It indicated that gelatin served as plasticizers to connect the zein nanoparticles and the electrospinnability was improved. Therefore, round and uniform nanofibers were observed.

The porosity results of the gelatin/zein nanofibers was shown in Figure 14. The porosities are between 60% to 70% except GZ21AA (22.5%) and GZ11AE (99.0%). Porosity is viewed as a crucial parameter for cellular infiltration and proliferation. The porosities of the gelatin/zein nanofibers were within a suitable range (60 - 90%) for cellular proliferation, which indicated its promising application in biomaterials.

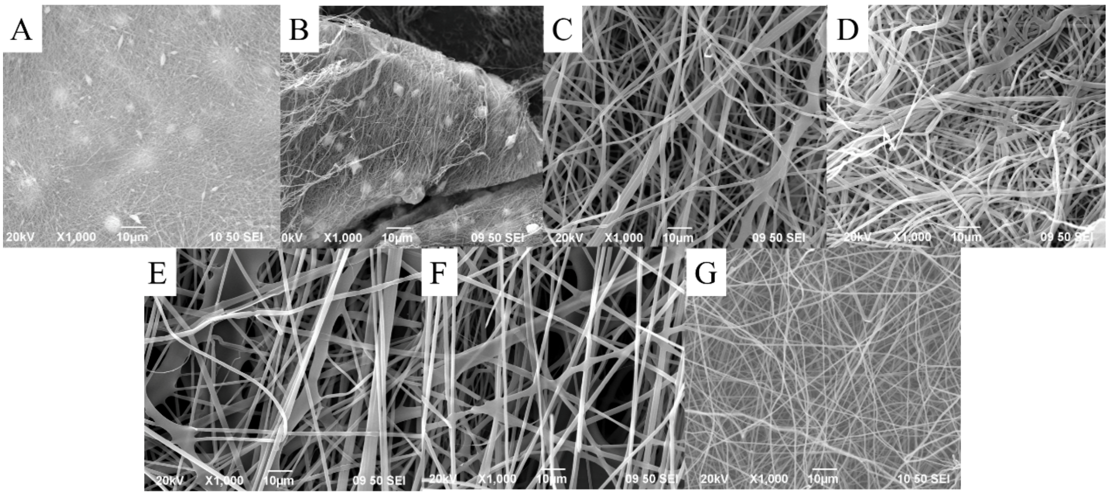


Figure 13. SEM images of gelatin/zein nanofibers in AA solution: (A) GZ01; (B) GZ13; (C) GZ12; (D) GZ11; (E) GZ21; (F) GZ31; (G) GZ10. (Scale bar = 10μm).

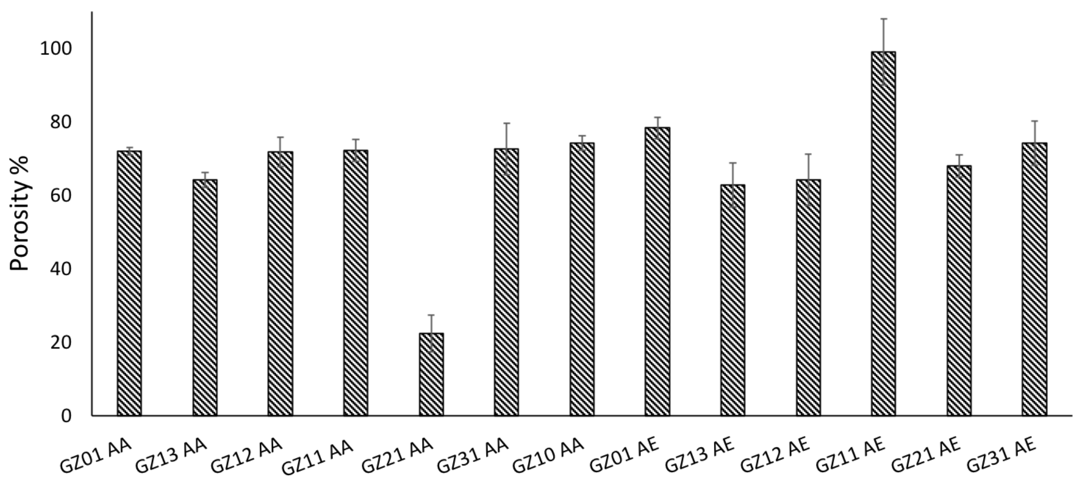


Figure 14. The porosities of gelatin/zein nanofibers.

To study the surface hydrophobicity of the gelatin/zein fibers, the measurements of water contact angles (WCAs) were conducted. Figure 15 showed the WCAs of gelatin/zein fibers fabricated using AA as solvent. The larger the WCA, the higher the hydrophobicity. When the gelatin/zein ratio is 1:1, the WCA is the highest (114.3°). To increase the content of either zein or gelatin, the WCA would decrease. When the content of gelatin was higher than 75% (GZ31 and GZ10), the WCAs were lower than 90° and the surfaces were hydrophilic. It was because gelatin was a water-soluble protein and a high content of gelatin will result in a hydrophilic surface. The surface of pure zein nanofibers (GZ01) had a contact angle of 103.6°. The reason was that the zein molecule was amphilic and it contained hydrophobic groups, which contributed on the formation of the surface of the nanofibers.

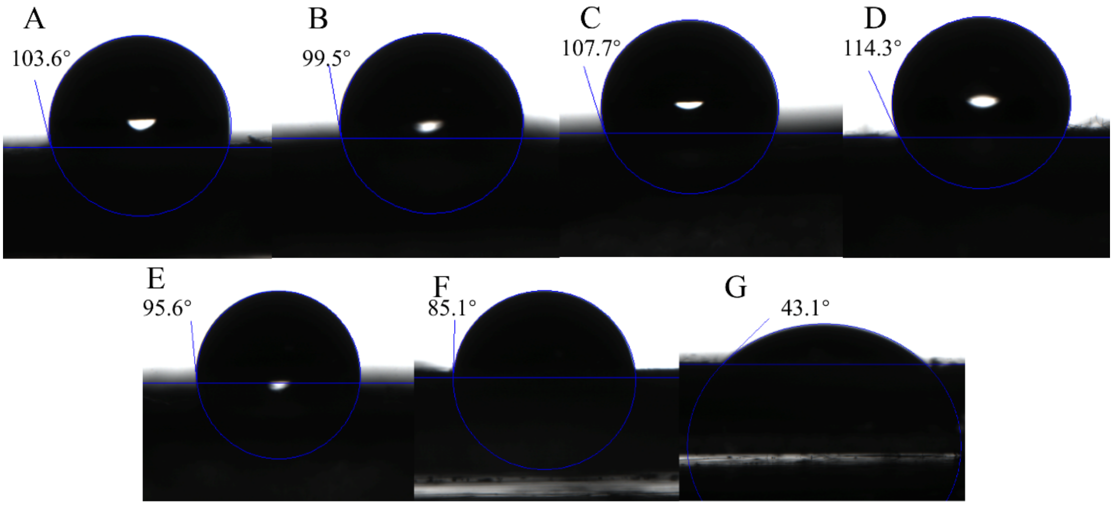


Figure 15. Water contact angles of gelatin/zein fibers in AA solution (A) GZ01; (B) GZ13; (C) GZ12; (D) GZ11; (E) GZ21; (F) GZ31; (G) GZ10.

Morphologies and hydrophobicities of gelatin/zein fibers fabricated using AE solution

The morphology of the electrospun gelatin/zein nanofibers using AE solution was shown in Figure 16. The viscosity of the pure gelatin solution (GZ10) is too high to form fibers. For pure zein fibers (GZ01), spindle-like fibers were formed with an average fiber diameter of 110.4 nm (Figure 16A). As gelatin added, round nanofibers with beads were produced (Figure 16B to C). For a further addition of gelatin to 50%

and higher (GZ11, GZ21, and GZ31), round and smooth nanofibers were produced without beads. The diameter of nanofibers were increased from 188.3 nm to 1650.5 nm as the weight ratio of gelatin increased from 25% (GZ13) to 66.7% （GZ21）, but a further addition of gelatin to 75% (GZ21) decreased the diameter to 844.0 nm. Comparing Figure 13 (A and B) and Figure 16 (A to C), the addition of ethanol might cause the assembly of zein and form more beads attached onto the fibers.

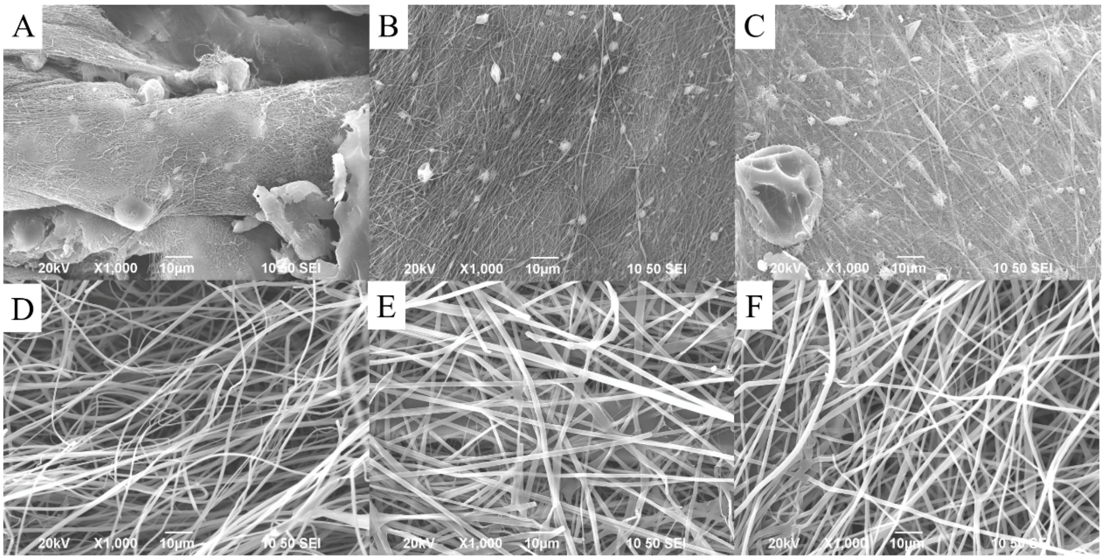


Figure 16. SEM images of gelatin/zein fibers in AE solution (A) GZ01; (B) GZ13; (C) GZ12; (D) GZ11; (E) GZ21; (F) GZ31(Scale bar = 10μm).

When using AE as the solvent, the water contact angles of nanofibers were shown in Figure 17. All the nanofibers were hydrophobic and showed comparable WCAs. With the addition of ethanol, the ratio of gelatin to zein has limited influence on the hydrophobic properties of nanofibers.

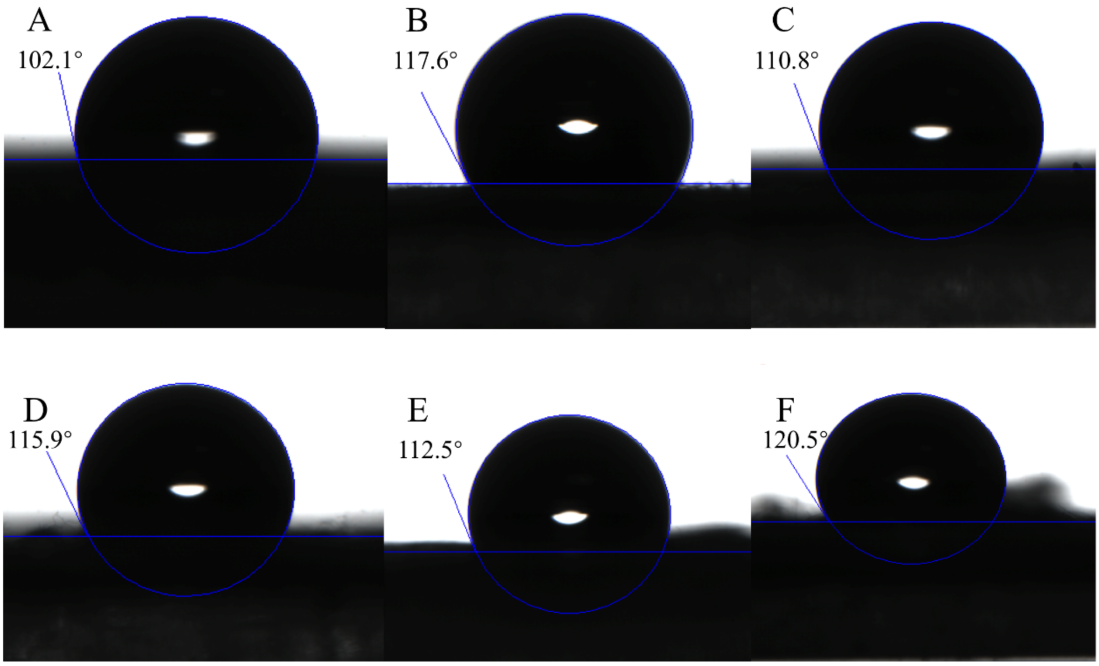


Figure 17. Water contact angles of gelatin/zein fibers in AE solution (A) GZ01; (B) GZ13; (C) GZ12; (D) GZ11; (E) GZ21; (F) GZ31.

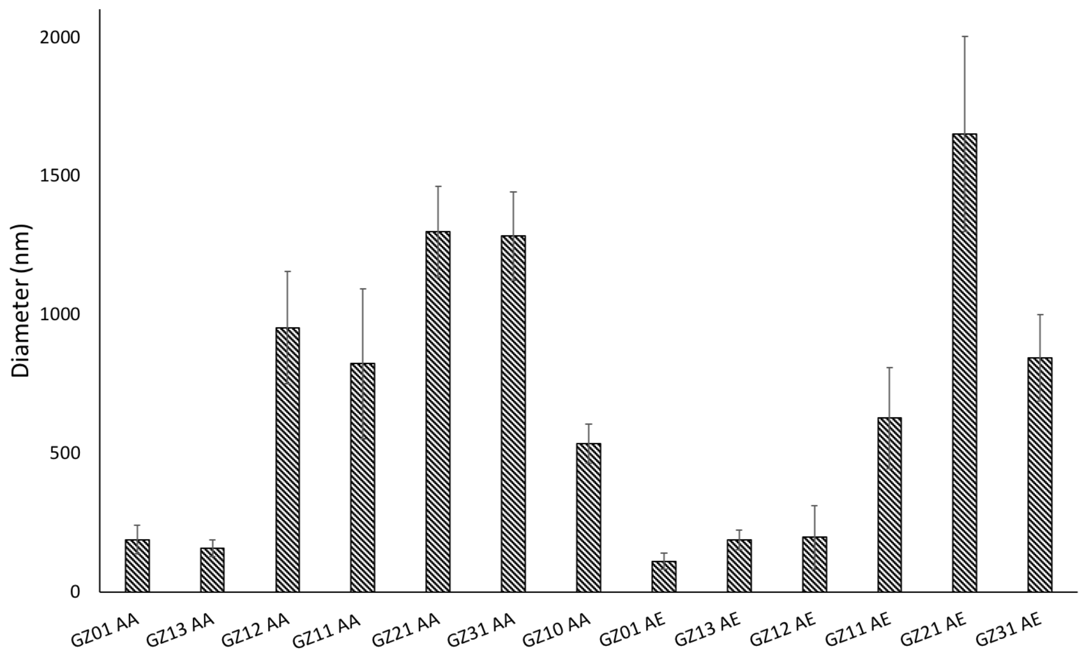


Figure 18. Diameter distribution of gelatin/zein nanofibers.

4.2.2 ATR-FTIR

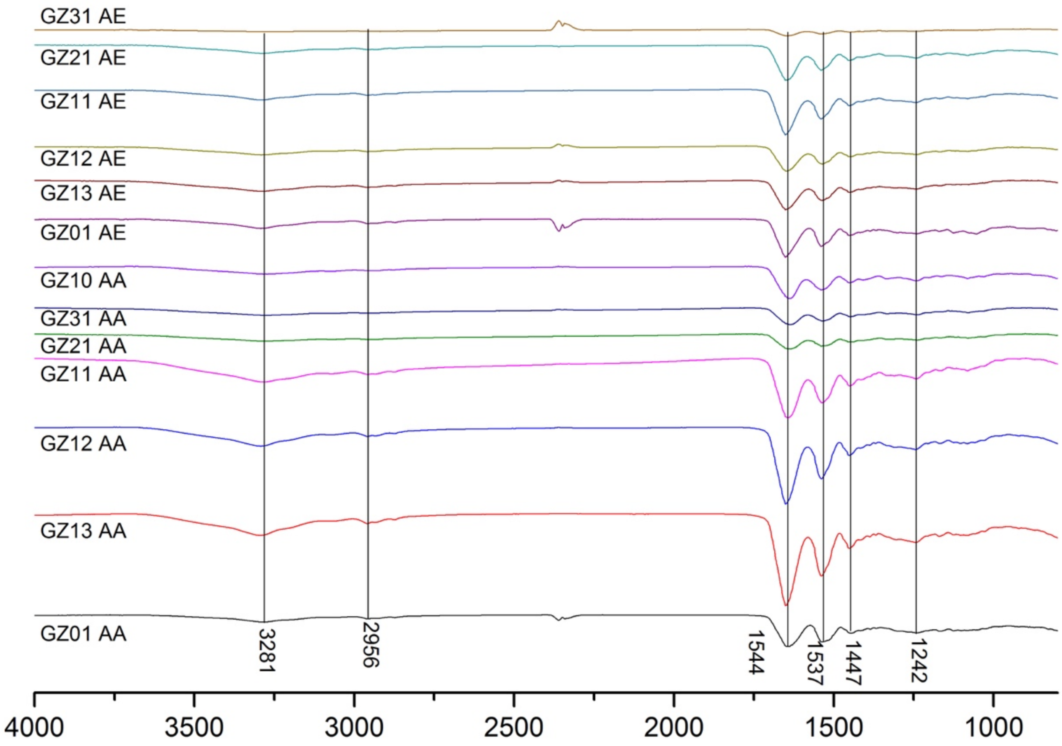
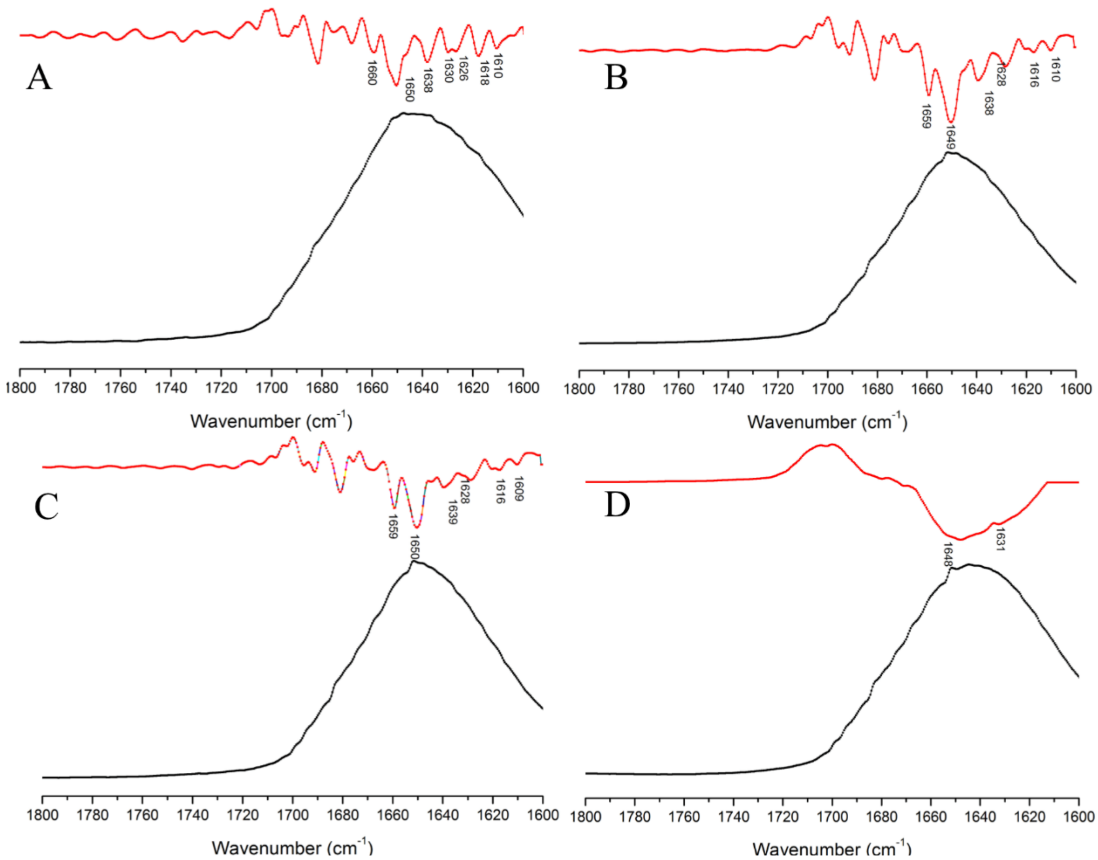


Figure 19. ATR- FTIR spectra of gelatin/zein fibers (AA=80% acetic acid, AE= acetic acid: ethanol: water =6:2:2), respectively.

To investigate the structures of nanofibers and possible interactions between the gelatin and zein molecules, ATR- FTIR was used and the spectra were obtained and shown in Figure 19. All samples showed typical bands of protein at approximately 1650 cm^{-1} (amide I) and 1540 cm^{-1} (amide II), corresponding to the stretching vibration of C=O bond and stretching of C-N bond, respectively. The absorption bands around 2960 cm^{-1} are attributed to the C-H stretching vibrations of the aliphatic groups. The absorption bands at 1447 cm^{-1} and 1242 cm^{-1} are attributed to the N-H bending and C-N stretching combination, and the C-N stretching and N-H bending combination, respectively. The broad absorption band between 3500 cm^{-1} and 3100 cm^{-1} is attributed to the contributions from amide A (N-H stretching vibration) (Morsy et al., 2017). There was no peak split in any of the hybrid nanofibers, indicating that gelatin and zein were uniformly dispersed within the fibers.



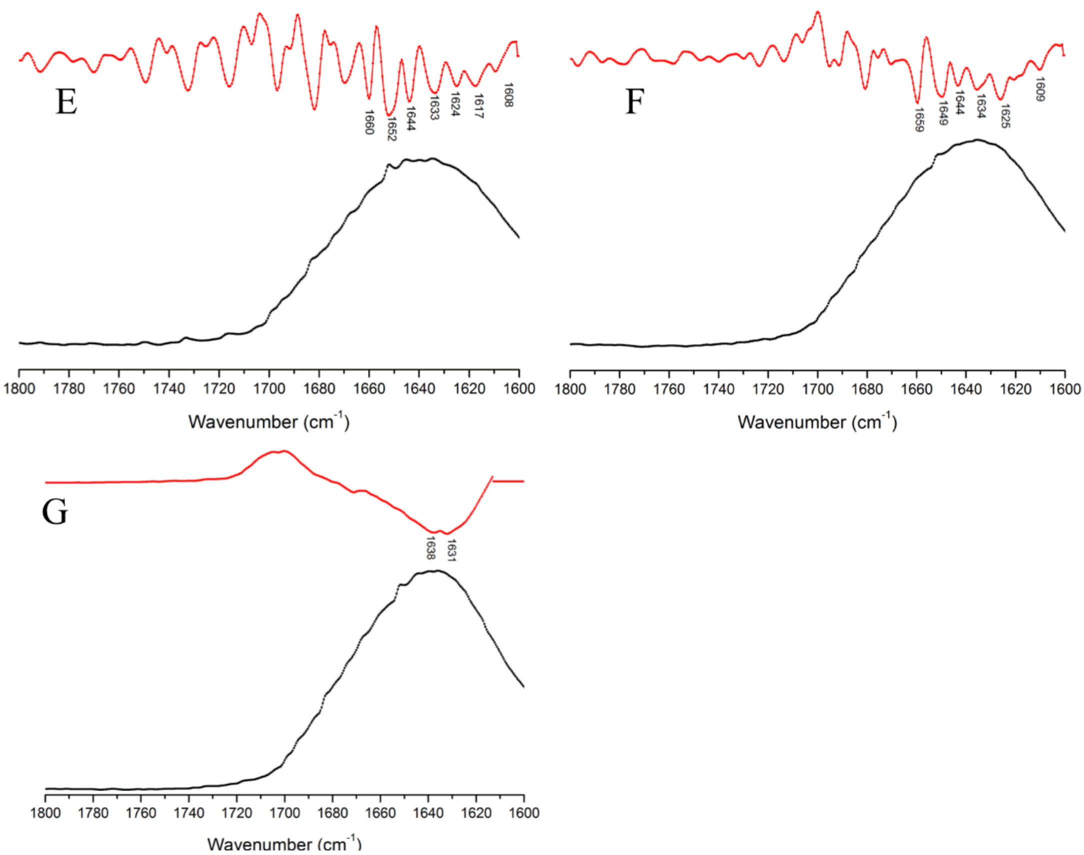


Figure 20. FTIR and 2th Derivation of gelatin/zein fibers in AA solution (A) GZ01; (B) GZ13; (C) GZ12; (D) GZ11; (E) GZ21; (F) GZ31; (G) GZ10.

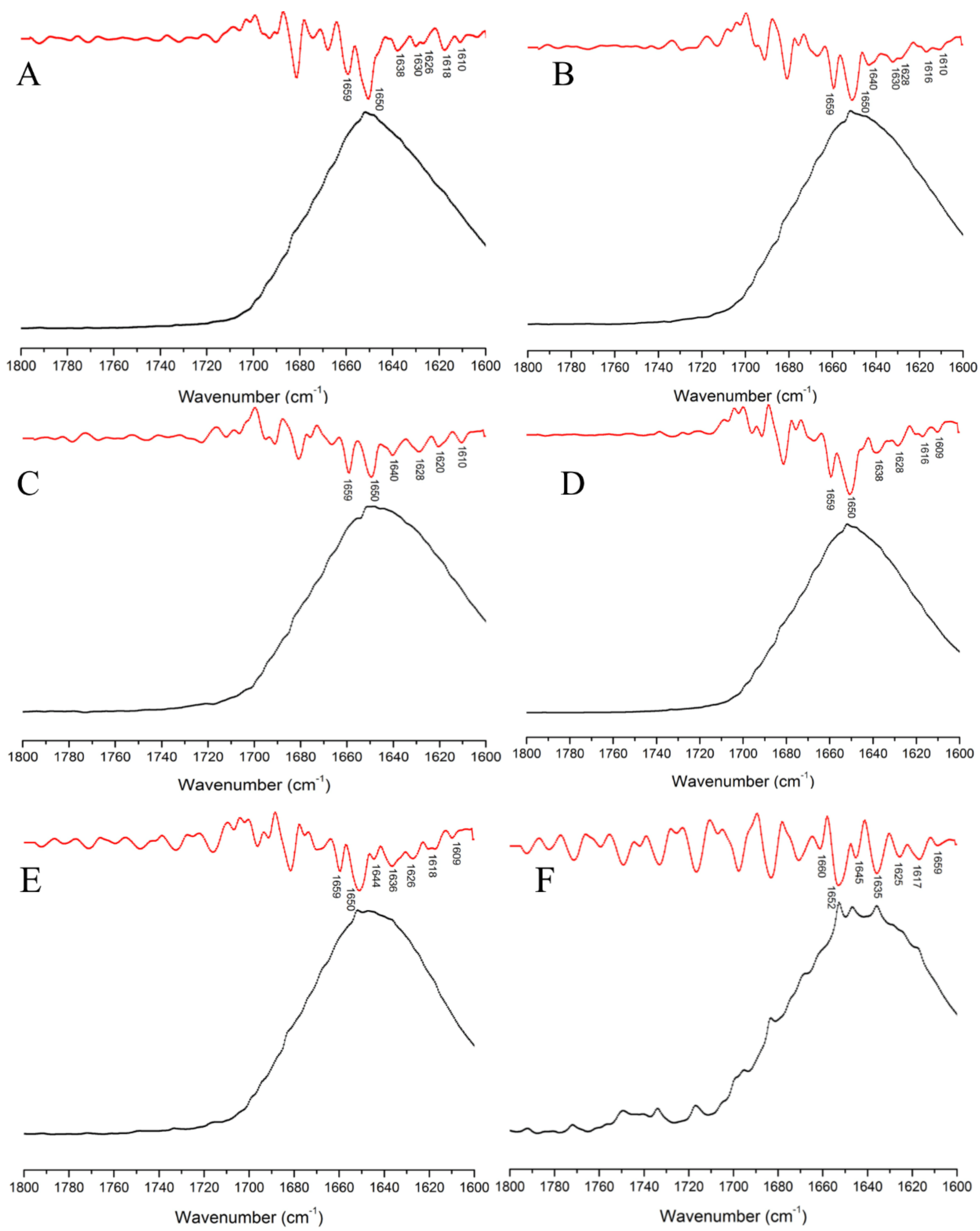


Figure 21. FTIR and 2th Derivation of gelatin/zein fibers in AE solution (A) GZ01; (B) GZ13; (C) GZ12; (D) GZ11; (E) GZ21; (F) GZ31.

Table 9. The ratio of the β-sheet content to α-helix content in the protein based nanofibers.

Sample	GZ01AA	GZ13AA	GZ12AA	GZ11AA	GZ21AA	GZ31AA	GZ10AA
Ratio	0.7±0.17	0.8±0.05	0.09±0.13	0.5±0.37	0.2±0.17	0.6±0.29	1.1±0.75
Sample	GZ01AE	GZ13AE	GZ12AE	GZ11AE	GZ21AE	GZ31AE	GZ10AE
Ratio	0.08±0.01	0.2±0.04	0.2±0.21	0.2±0.09	0.3±0.01	0.3±0.11	NA

4.2.3 The transformation of α-helix and β-sheet

To further study the secondary structure of the protein in the nanofibers, the ATR-FTIR spectra were collected and the corresponding secondary derivatives were presented in Figures 20 and 21. As shown, the spectra of all the samples had a peak at 1650 cm^{-1}, which is attributed to the α-helix structure, while the peak at 1628 cm^{-1} indicated the β-sheet structure. To study the transformation of α-helix and β-sheet, the intensity ratio of peaks 1628 cm^{-1} to 1650 cm^{-1} was calculated to show the ratio of the β-sheet content to α-helix content in the protein based nanofibers (Table 9). As shown, when the solvent was AA, the content of β-sheet structure was first decreased and then increased as the content of zein increased from 0 to 100% in the zein-gelatin mixture. However, when using AE as the solvent, as the weight ratio of gelatin and zein changed, the content of β-sheet was maintained in a relatively low level.

4.2.4 Mechanical strength

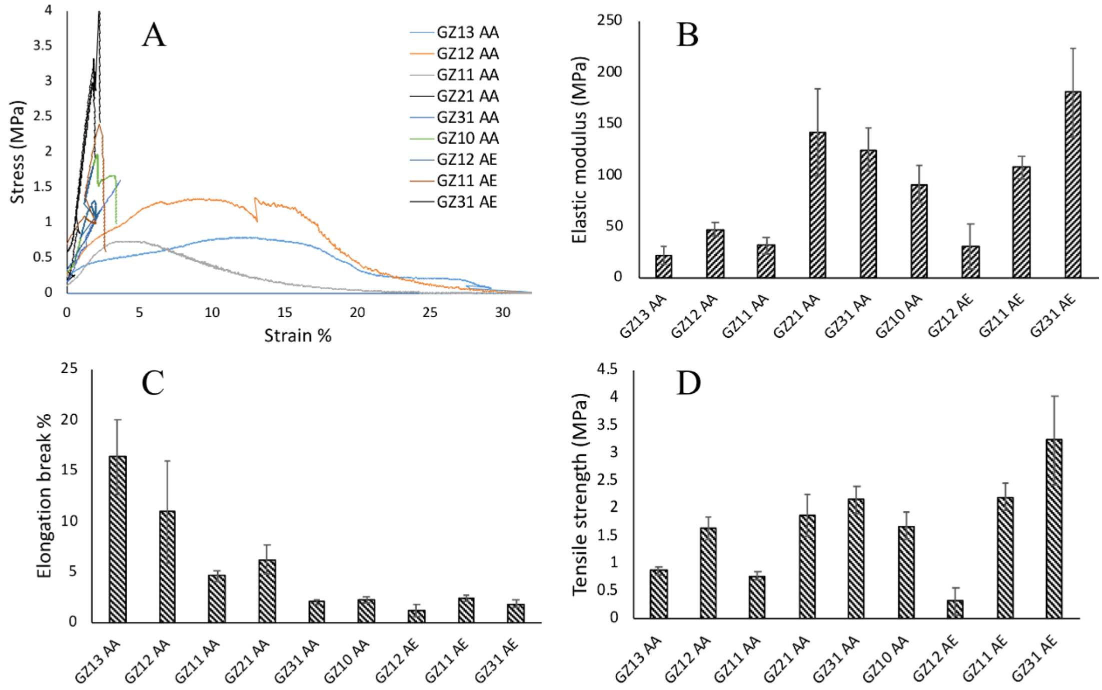


Figure 22: Mechanical properties of the gelatin/zein nanofibers in dry conditions: (A) typical stress-strain curves; (B) elastic modulus; (C) elongation at break; (D) tensile strength.

As we observed that, during oil absorption, the nanofiber network will expand. And because the volume increases, more oil can be absorbed. To evaluate the expansion degree of the network, the mechanical strength of the nanofibers were measured. The typical stress-strain curves and the calculated elastic modulus, elongation at break, and tensile strength of the gelatin/zein nanofibers were measured and shown in Figure 22. It can be observed that the nanofibers GZ01AA, GZ01AE, GZ13AE, and GZ21AE were too weak and no the mechanical strength results were obtained. The GZ31AE nanofibers showed the highest elastic modulus of 180.9 MPa (Figure 12B). As shown in Figure 22C, the nanofibers fabricated using AE as the solvent showed very low elongation (less than 3%). And with AA as the fabrication solvent, higher ratios of zein, GZ13AA and GZ12AA, gave rise to a much higher elongation--at-break of 16.4% and 11.0%, which indicated good deformability and flexibility. Once the zein content was decreased to 50% (GZ11AA), the nanofiber become fragile with an elongation-at-break of 4.7%. With AE as the fabrication solvent (GZ12AE, GZ11AE, and GZ31AE), the tensile strength increased with the increasing content of gelatin (Figure. 22D).

Weak interactions between zein particles hold a continuous network of the compact particles through a gelatin matrix, which leads to a faster and more uniform diffusion of the stress. This phenomenon probably affects the mechanical properties of the composites, as previously reported in the zein hordein system (Wang and Chen, 2012).

Figure 22C, with the addition of gelatin, the elongation-at-break is reduced when AA was used as the fabrication solvent. The elongation at break of zein film (at 25^{o}C and 50% RH) typically is 12% (Lai and Padua, 2012), while that of gelatin film is 4.2% (de Carvalho and Grosso, 2004). The decrease of elongation-at-break is caused by the addition of gelatin.

Figure 22D, with the addition of gelatin, the tensile strength is increased when AE was used as the fabrication solvent. Figure 22D, with the addition of gelatin, the tensile strength is increased when AE was used as the fabrication solvent. The tensile strength of zein film (80% ethanol) generally is 9 MPa (at 25 ^{o}C and 50% RH), while that of gelatin film is 15.1 MPa (de Carvalho and Grosso, 2004). The addition of gelatin enhanced the tensile strength of composite nanofibers. This phenomenon was probably attributed to immiscibility and separation of microphase. It could be concluded that the formation of the networks assembled with particles of zein uniformly dispersed in a gelatious matrix provided a strategy to improve the stability and mechanical strength of electrospun protein fibers, which might endow them wider applications.

4.2.5 Thermal analysis

TGA analysis was performed to investigate the degradation of the gelatin/zein nanofibers at various weight ratios and the result was shown in Figure 23. It was said that the thermal degradation of biopolymers usually involved two steps: the initial stage was related to the degradation of biopolymer network, followed by the degradation of the inner covalent bonds of biopolymer monomers. The TGA curves of the gelatin/zein fibers showed two zones of weight loss. The presence of a residual mass at temperatures higher than 500 °C was ascribed to inorganic compounds derived from thermal degradation. Therefore, the increased residue with the increasing weight percentage of gelatin in the blend was attributed to the higher ratio of inorganic compound in gelatin.

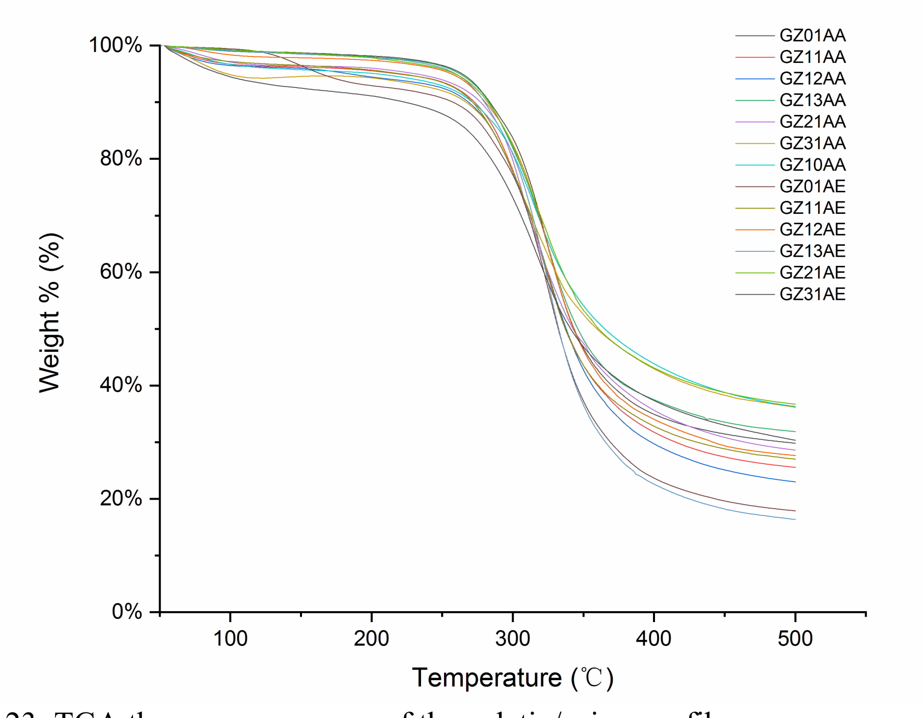


Figure 23: TGA thermogram curves of the gelatin/zein nanofibers.

4.2.6 Oil absorption ability

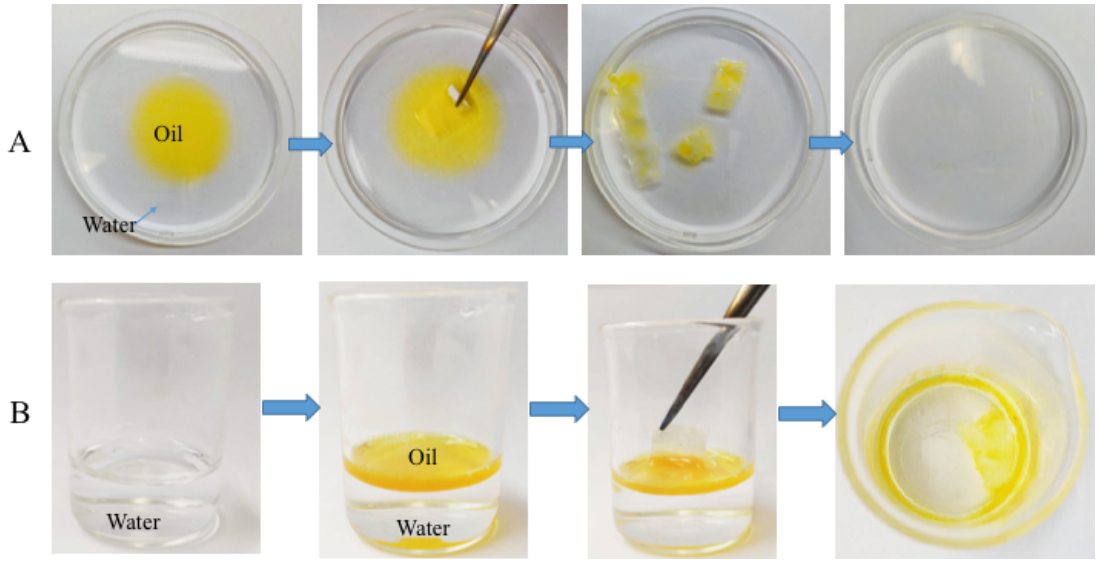


Figure 24. Optical images show the process of oil removal (olive oil) with nanofibers from the water/oil mixture at different intervals.

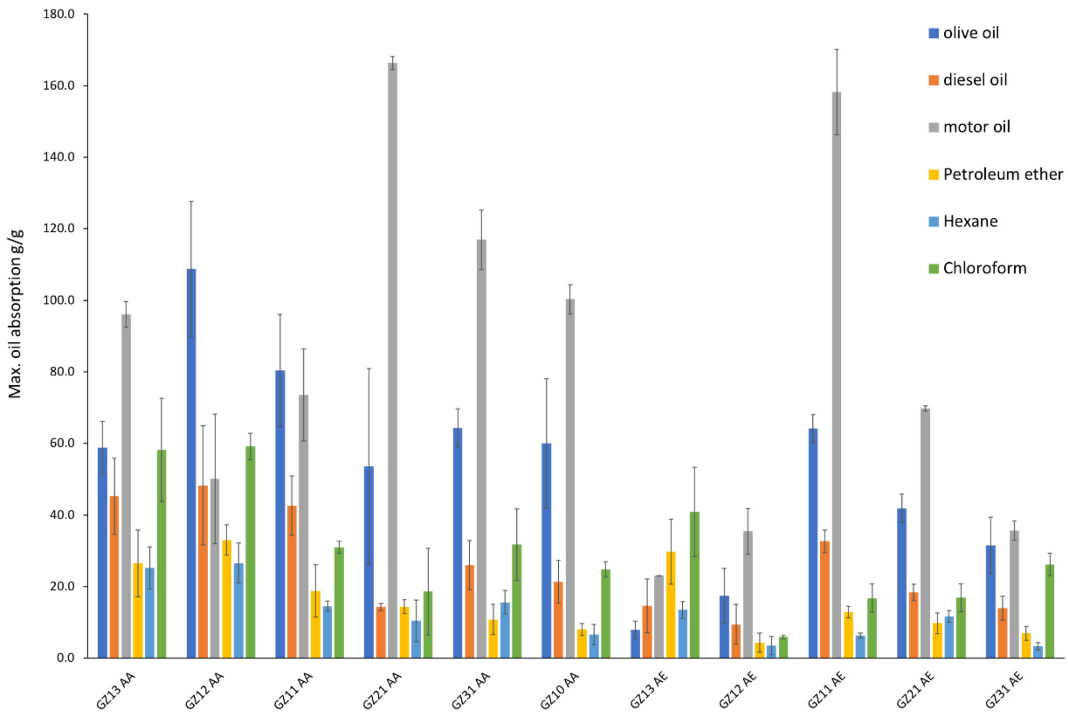


Figure 25. Maximum absorption capacities of various oils, including motor oil, diesel oil, olive oil, petroleum ether, n-hexane and chloroform onto the gelatin/zein fibers at

25°C.

To investigate the absorption performance of the nanofibers, the olive oil was dyed and an oil/water mixture was formed. Then nanofiber sample was put into the water/oil system and pictures at various time points during the oil cleanup process was recorded and shown in Figure 24. It was observed that the nanofibers turned yellow after absorbing the olive oil, whereas water become colorless after the oil absorption by nanofibers. It indicated that the gelatin/zein nanofibers have excellent affinity for absorption of the oil and can selectively remove oil from the water/oil mixture.

Further, to measure the oil absorption ability of the nanofibers, various oils and organic solvents, including olive oil, diesel oil, motor oil, petroleum ether, n-hexane, and chloroform, were selected. The related physical properties of the oils were listed in Table 12 and the oil absorption results were shown in Figure 11. For example, GZ13AA has the maximum absorption capacity of 58.9 g/g for olive oil, 45.3 g/g for diesel oil, 96.1 g/g for motor oil, 26.5 g/g for petroleum ether, 25.2 g/g for n-hexane, and 58.2 g/g for chloroform. Generally, for the same oil and with the same ratio of zein and gelatin, the nanofibers prepared using AA solvent have a larger absorption capacity than that using AE solvent. According to Figures 18 and 25, the nanofibers with higher oil absorption capacity usually had larger diameter. The nanofibers with larger diameter, to a certain degree, can expand the network. And more voids or interspace, where the oil is trapped during the adsorption, are accordingly created, enhancing the absorption capacity. The high adsorption capacity of the fibrous film is the combined results of oleophilicity caused by the low surface tension and the voids between the interconnected fibers. At the initial stage, the oil adsorption strongly depends on the oleophilicity, for example, the chemical compatibility between the oil and the nanofibers. For the porous nanofibers, the absorption capacity is mainly determined by the surface tension and viscosity of the oil. The lower the surface tension of the oil is, the more easily the nanofiber is wetted by the oil. In addition, because of the capillary force, the porous structure promotes the oil diffusion into the membrane. Once the nanofiber is wetted, the oil viscosity would exert an important influence on the absorption capacity. The higher viscosity can increase the absorption capacity by improving the adherence of oil onto the fiber surface, although the rate of the oil diffusion and penetration resulting from the capillary action is inversely proportional to the oil viscosity.

Based on the results shown in Figure 25 and Table 10, the oil viscosity, compared with the oil surface tension, plays a more important role on determining the absorption capacity. It can be found that the absorption capacity of the motor oil is larger than that of other oils. Take the GZ13AA as an example, among the oils and organic solvents, the absorption capacity increases as: diesel oil < olive oil < motor oil, and similar trend is also found for the other nanofibers, except GZ12AA and GZ11AA.

Table 10. Physical properties of oils and organic solvents used at room temperature (25 °C) (Ragab et al., 2015; Sahasrabudhe et al., 2017; Schaschke et al., 2013; Hersh et al., 1935; Michailidou et al., 2013 Clara et al., 2010).

Oil	Viscosity (mPa S)	Density (g cm^{-1})	Surface tension (μN m^{-1})
Motor oil	290	0.89	23.53
Olive oil	74.1	0.91	31.9
Diesel oil	50	0.72	24.39
Petroleum ether	0.3	0.64-0.66	17.5
n-Hexane	0.33	0.66	19.04
Chloroform	0.54	1.49	26.67

4.2.7 Multifunctional applications

It's a challenge to remove oil from emulsified oil/water mixtures in oily sewage treatment due to the small drop size and the low interfacial energy. The ability of nanofibrous membrane removing micrometer-sized oil droplets was investigated. A toluene-in -water emulsion (1:9, v/v) with a droplet size at the micrometer scale was prepared. As shown in Figure 26a the emulsion was a white milky liquid before separation. It turned to transparent quickly when the nanofibrous membrane was put into the mixture. For clearly observing the separation, the optical microscopic images of the emulsion before and after separation were obtained and shown in Figure 12b and c. Figure 26b shows that there were large amounts of droplets in the feeds of emulsions before separation. Almost no oil droplets were observed in the liquid after separation (Figure 26c). It indicates that almost all of toluene in the emulsion was eliminated and the nanofibrous membrane was in high potential for the separation of oil-in-water emulsions.

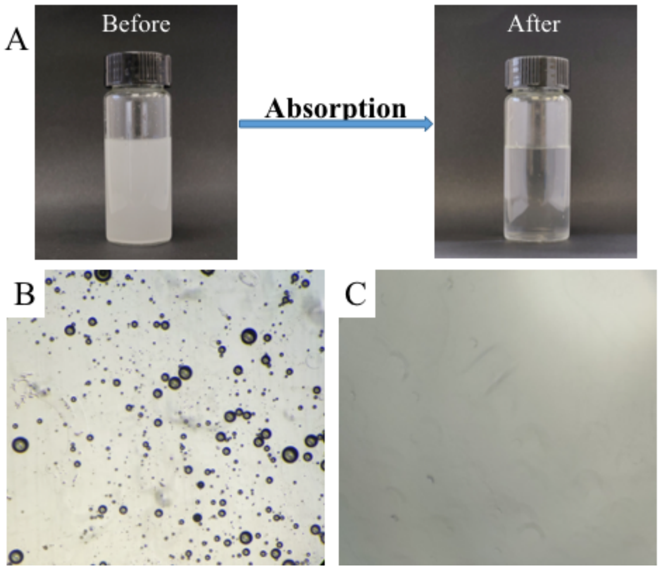


Figure 12. Optical images of toluene-in-water emulsion before and after nanofibrous membrane separation (A) optical micrographs of the toluene-in-water emulsion before (B) and after (C) nanofibrous membrane separation.

Generally, heavy metals are often found in wastewater and it's also hard to clean. To confirm that the nanofibrous membrane could also absorb heavy metals from wastewater by physical adsorption, EDS data of membrane after Cu^{2+} absorption were taken at spot 1 (Figure 27 A and C) and area 2 (Figure 27 B and D). And the relative atomic percentages of Cu, C and O at spot 1 and area 2 in nanofibrous membrane were shown in Table 11. The determined atomic ratio of Cu for spot 1 is 3.64 and area 2 is 3.10. As the results shown, after 24h absorption, the element of Cu was appeared which means it could be adsorbed by nanofibrous membrane.

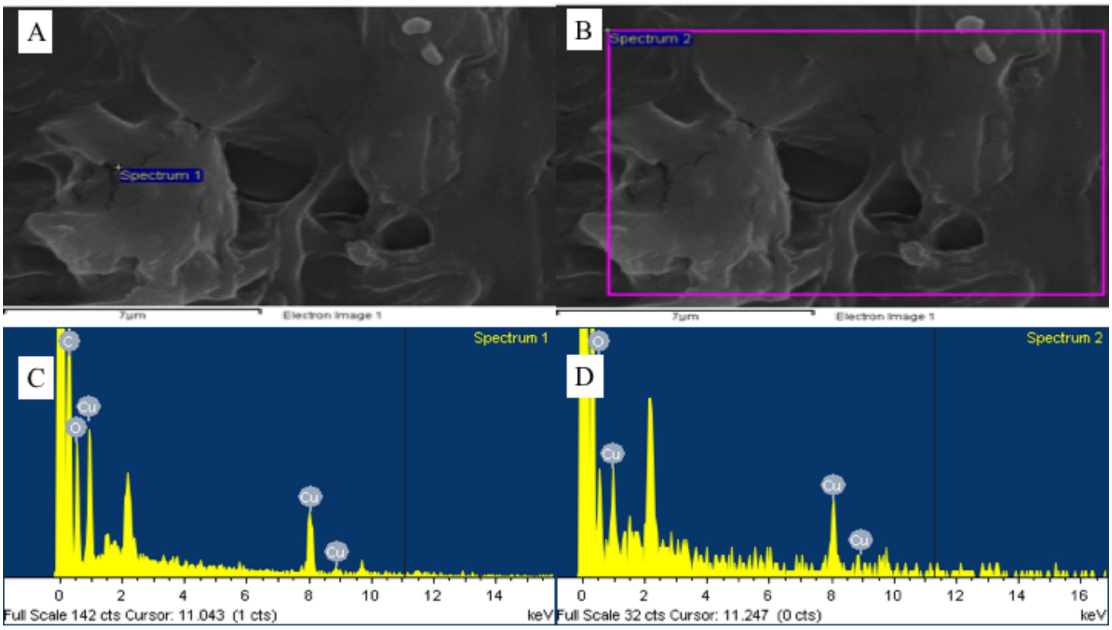


Figure 27. SEM images of nanofibers after Cu^{2+} absorption where EDS data were taken and EDS maps. (A) and (B) SEM images where EDS data were taken at spot 1 and area 2; (C) and (D) EDS maps of spot 1 and area 2.

Table 11. The relative atomic percentage of Cu, C and O at spot 1 and area 2 in the nanofibrous membrane

Element	**Atomic percentage %**	
	Spot 1	**Area 2**
C	68.65	72.72
O	27.71	24.19
Cu	3.64	3.10
Totals	100.00	100.00

4.3 Conclusion

The gelatin/zein nanofibers were fabricated by hybrid electrospinning and obtained hydrophobic nanofibers with high oil absorption capacity. The electrospinning process dispersed gelatin well into zein network to form a uniform nanofiber, resulting in a good thermal stability and hydrophobicity. The mechanical tests showed that the gelatin/zein nanofibers in 80% acetic acid with weight ratios of 1:3 and 1:2 gave rise to much higher elongation at break of 16.4% and 11.0%, indicating good deformability and flexibility. The nanofibers showed extremely high absorbing abilities for organic liquids with the highest oil absorption 109 g/g for olive oil, 48 g/g for diesel oil and 166 g/g for motor oil, approximately 4-6 times that of natural sorbents and nonwoven polypropylene fibrous mats. It suggested that the hybrid electrospun gelatin/zein nanofibers had a potential promising application for oil water separation.

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